

Introduction to Algebra I (revisit)

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1. Groups I

Lecture 1: Permutations, Groups, Lagrange's Theorem

§1.1. Permutations

Rotman
§2.2–§2.4

In high school mathematics the words *permutation* and *arrangement* are used interchangeably. We draw a distinction between them.

Definition 1.1.1 (Permutation, arrangement). A **permutation** of a set X is a bijection $\alpha: X \rightarrow X$. If X is finite with $|X| = n$, then an **arrangement** of X is a list $x_1, x_2, \dots, x_n$ of all the elements of X with no repetitions.

Given an arrangement $x_1, \dots, x_n$, define $f: \{1, \dots, n\} \rightarrow X$ by $f(i) = x_i$; thus the list displays the values of f . That there are no repetitions says f is injective, and that every $x \in X$ occurs on the list says f is surjective. So an arrangement of X is the same thing as a bijection $f: \{1, \dots, n\} \rightarrow X$.

Example 1.1.2 (Arrangements of a three-element set). There are six arrangements of $X = \{a, b, c\}$:

$$abc; \quad acb; \quad bac; \quad bca; \quad cab; \quad cba.$$

In general there are exactly $n!$ arrangements of an n -element set.

If $X = \{1, 2, \dots, n\}$, a permutation $\alpha: X \rightarrow X$ gives the list $\alpha(1) = i_1, \dots, \alpha(n) = i_n$, which we record in **two-rowed notation**:

$$\alpha = \begin{pmatrix} 1 & 2 & \cdots & j & \cdots & n \\ \alpha(1) & \alpha(2) & \cdots & \alpha(j) & \cdots & \alpha(n) \end{pmatrix}.$$

Intuition . Informally, arrangements (lists) and permutations (bijections) are two ways of describing the same thing. The advantage of viewing a permutation as a *bijection* is that bijections can be *composed*, and the composite of two permutations is again a permutation. Lists can only be counted.

Definition 1.1.3 (Symmetric group). The family of all permutations of a set X , denoted by S_X , is called the **symmetric group** on X . When $X = \{1, 2, \dots, n\}$ we write S_n and call it the **symmetric group on n letters**.

Example 1.1.4 (Composition in S_3 is not commutative). If

$$\alpha = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \quad \text{and} \quad \beta = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix},$$

then

$$\alpha \circ \beta = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \quad \text{and} \quad \beta \circ \alpha = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix},$$

so $\alpha \circ \beta \neq \beta \circ \alpha$: for instance $\alpha \circ \beta: 1 \mapsto \alpha(\beta(1)) = \alpha(2) = 3$, whereas $\beta \circ \alpha: 1 \mapsto 2 \mapsto 1$.

Some permutations do commute, however; the reader may check that $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 3 & 4 \end{pmatrix}$ and $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix}$ commute.

Note 1.1.5 (Order of multiplication). Some authors put “functions on the right”, writing $(i)\alpha$ instead of $\alpha(i)$. Their $\alpha \circ \beta$ is our $\beta \circ \alpha$: the notational switch causes a switch in the order of multiplication.

Proposition 1.1.6 (Cancellation in S_X)

If $\gamma, \alpha, \beta \in S_X$ and $\gamma\alpha = \gamma\beta$, then $\alpha = \beta$. Similarly $\alpha\gamma = \beta\gamma$ implies $\alpha = \beta$.

Proof. Compose with γ^{-1} and use associativity of composition:

$$\alpha = 1_X \circ \alpha = (\gamma^{-1} \circ \gamma) \circ \alpha = \gamma^{-1} \circ (\gamma \circ \alpha) = \gamma^{-1} \circ (\gamma \circ \beta) = (\gamma^{-1} \circ \gamma) \circ \beta = 1_X \circ \beta = \beta.$$

A similar argument works on the other side. ■

Aside from being cumbersome, the two-rowed notation hides the answers to elementary questions: is the square of a permutation the identity? what is the smallest $m > 0$ with $\alpha^m = 1$? can one factor a permutation into simpler pieces? The special permutations below remedy this defect.

Notation . From now on we write $\beta\alpha$ instead of $\beta \circ \alpha$, and (1) instead of 1_X .

§1.1.i. Cycles

Definition 1.1.7 (Fixes, moves). If $\alpha \in S_n$ and $i \in \{1, \dots, n\}$, then α **fixes** i if $\alpha(i) = i$, and α **moves** i if $\alpha(i) \neq i$.

Definition 1.1.8 (Cycle). Let $i_1, i_2, \dots, i_r$ be distinct integers in $\{1, \dots, n\}$. If $\alpha \in S_n$ fixes the other integers (if any) and

$$\alpha(i_1) = i_2, \quad \alpha(i_2) = i_3, \quad \dots, \quad \alpha(i_{r-1}) = i_r, \quad \alpha(i_r) = i_1,$$

then α is called an **r -cycle**, or a **cycle of length r** , and we write

$$\alpha = (i_1 \ i_2 \ \dots \ i_r).$$

A 2-cycle is called a **transposition**. A 1-cycle is the identity, so all 1-cycles are equal: $(i) = (1)$ for every i .

Example 1.1.9 (Cycle notation). The two-rowed notation does not help us recognize that $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 3 & 1 & 5 & 2 \end{pmatrix}$ is a 5-cycle, but in fact $\alpha(1) = 4$, $\alpha(4) = 5$, $\alpha(5) = 2$, $\alpha(2) = 3$, $\alpha(3) = 1$; that is, $\alpha = (1 \ 4 \ 5 \ 2 \ 3)$. Likewise

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} = (1 \ 2 \ 3 \ 4), \quad \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 1 & 4 & 2 & 3 \end{pmatrix} = (1 \ 5 \ 3 \ 4 \ 2),$$

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 4 & 5 \end{pmatrix} = (1 \ 2 \ 3).$$

On the other hand $\beta = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix}$ is *not* a cycle; in fact $\beta = (1 \ 2)(3 \ 4)$.

The term *cycle* comes from the Greek word for circle: one pictures the cycle $(i_1 \ i_2 \ \dots \ i_r)$ as a clockwise rotation of a circle (Figure 1.1). Any i_j may be taken as the starting point, so an r -cycle has r different cycle notations:

$$(i_1 \ i_2 \ \dots \ i_r) = (i_2 \ i_3 \ \dots \ i_r \ i_1) = \dots = (i_r \ i_1 \ i_2 \ \dots \ i_{r-1}).$$

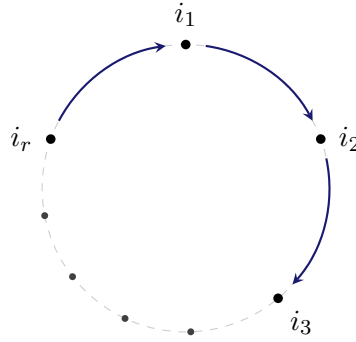


Figure 1.1.: A cycle as a (clockwise) rotation.

Example 1.1.10 (The factorization algorithm). Take

$$\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 6 & 4 & 7 & 2 & 5 & 1 & 8 & 9 & 3 \end{pmatrix}.$$

Begin by writing “(1”. Now $\alpha: 1 \mapsto 6$, so write “(1 6”. Next $\alpha: 6 \mapsto 1$, so the parentheses close: α begins “(1 6)”. The smallest number not yet used is 2, and $2 \mapsto 4$, $4 \mapsto 2$, giving “(1 6)(2 4)”. The smallest remaining number is 3, and $3 \mapsto 7 \mapsto 8 \mapsto 9 \mapsto 3$, giving the 4-cycle (3 7 8 9). Finally $\alpha(5) = 5$, and so

$$\alpha = (1\ 6)(2\ 4)(3\ 7\ 8\ 9)(5).$$

To verify a claim like this one evaluates both sides. For instance, writing $\beta = (1\ 6)$, $\gamma = (2\ 4)$, $\delta = (3\ 7\ 8\ 9)$,

$$\beta\gamma\delta(1) = \beta(\gamma(\delta(1))) = \beta(\gamma(1)) = \beta(1) = 6 = \alpha(1),$$

since δ and γ fix 1. ([Proposition 1.1.16](#) gives a more satisfactory proof that the algorithm works.)

Example 1.1.11 (Multiplying with the algorithm). Factorizations into cycles are convenient for *multiplying* permutations: one simply displays the partial outputs. In S_5 let

$$\sigma = (1\ 2)(1\ 3\ 4\ 2\ 5)(2\ 5\ 1\ 3).$$

Then $\sigma: 1 \mapsto 3 \mapsto 4 \mapsto 2 \mapsto 1$, so σ begins “(1 4”. Next $4 \mapsto 2 \mapsto 1$, so σ begins “(1 4)”. Then $2 \mapsto 5 \mapsto 1 \mapsto 2$, so σ fixes 2; and $3 \mapsto 2 \mapsto 5 \mapsto 3$, $5 \mapsto 1 \mapsto 3 \mapsto 5$. Therefore $\sigma = (1\ 4)(2)(3\ 5)$.

Definition 1.1.12 (Disjoint permutations). Two permutations $\alpha, \beta \in S_n$ are **disjoint** if every i moved by one is fixed by the other: if $\alpha(i) \neq i$ then $\beta(i) = i$, and if $\beta(j) \neq j$ then $\alpha(j) = j$. A family $\beta_1, \dots, \beta_t$ is disjoint if each pair of them is disjoint.

Observation . If $\alpha = (i_1 \dots i_r)$ and $\beta = (j_1 \dots j_s)$, then any $k \in \{i_1, \dots, i_r\} \cap \{j_1, \dots, j_s\}$ is moved by both. Hence two cycles are disjoint if and only if the underlying sets $\{i_1, \dots, i_r\}$ and $\{j_1, \dots, j_s\}$ are disjoint sets. When α and β are disjoint there are exactly three possibilities for a number i : it is moved by α , it is moved by β , or it is fixed by both.

Lemma 1.1.13 (Disjoint permutations commute)

Disjoint permutations $\alpha, \beta \in S_n$ commute.

Proof. It suffices to prove $\alpha\beta(i) = \beta\alpha(i)$ for $1 \leq i \leq n$. If β moves i , say $\beta(i) = j \neq i$, then β also moves j (otherwise $\beta(j) = j$ and $\beta(i) = j$ contradict injectivity of β); since α and β are disjoint, $\alpha(i) = i$ and $\alpha(j) = j$, so $\beta\alpha(i) = \beta(i) = j = \alpha(j) = \alpha\beta(i)$. A similar argument applies if α moves i . The last possibility is that neither moves i , and then $\alpha\beta(i) = i = \beta\alpha(i)$. ■

Remark 1.1.14. The converse fails: permutations that are *not* disjoint may still commute. The reader may check that $(1\ 2\ 3)(4\ 5)$ and $(1\ 3\ 2)(6\ 7)$ commute; even simpler, $\alpha\alpha^2 = \alpha^2\alpha$.

Lemma 1.1.15

Let $X = \{1, \dots, n\}$, let $\alpha \in S_X = S_n$, and given $i_1 \in X$ define $i_{j+1} = \alpha(i_j)$ for $j \geq 1$. Write $Y = \{i_j : j \geq 1\}$ and let Y' be the complement of Y in X .

- (i) If α moves i_1 , then there is $r > 1$ with $i_1, \dots, i_r$ all distinct and $i_{r+1} = \alpha(i_r) = i_1$.
- (ii) $\alpha(Y) = Y$ and $\alpha(Y') = Y'$.

Proof. (i) Since X is finite, there is a smallest $r > 1$ with $i_1, \dots, i_r$ all distinct but $i_{r+1} = \alpha(i_r) \in \{i_1, \dots, i_r\}$; say $\alpha(i_r) = i_j$ with $1 \leq j \leq r$. If $j > 1$, then $\alpha(i_r) = i_j = \alpha(i_{j-1})$, and injectivity of α gives $i_r = i_{j-1}$, contradicting distinctness. Therefore $\alpha(i_r) = i_1$.

(ii) Clearly $\alpha(Y) \subseteq Y$, since $i_j \in Y$ implies $\alpha(i_j) = i_{j+1} \in Y$. If $k \in Y'$ and $\alpha(k) \in Y$, then $\alpha(k) = i_j = \alpha(i_{j-1})$ for some j (by part (i) this holds even for $i_j = i_1$); injectivity gives $k = i_{j-1} \in Y$, contradicting $Y \cap Y' = \emptyset$. Hence $\alpha(Y') \subseteq Y'$.

These inclusions are equalities: $\alpha(X) = \alpha(Y) \cup \alpha(Y')$ is a disjoint union because α is injective, and $|\alpha(Y)| \leq |Y|$, $|\alpha(Y')| \leq |Y'|$. If either inequality were strict, then $|\alpha(X)| < |X|$, contradicting surjectivity of α . ■

Proposition 1.1.16 (Factorization into disjoint cycles)

Every permutation $\alpha \in S_n$ is either a cycle or a product of disjoint cycles.

Proof. Induct on the number $k \geq 0$ of points moved by α . The base step $k = 0$ is true, for then α is the identity, which is a 1-cycle.

If $k > 0$, let i_1 be a point moved by α . As in Lemma 1.1.15, define $Y = \{i_1, \dots, i_r\}$ with the i_j distinct, $\alpha(i_j) = i_{j+1}$ for $j < r$, and $\alpha(i_r) = i_1$. Let $\sigma = (i_1\ i_2\ \dots\ i_r)$, so that σ fixes each point of Y' . If $r = n$, then $\alpha = \sigma$. If $r < n$, then $\alpha(Y') = Y'$ by the lemma. Put $\alpha' = \alpha\sigma^{-1}$; we claim α' and σ are disjoint. If σ moves i , then $i = i_j \in Y$, and $\alpha'(i_j) = \alpha\sigma^{-1}(i_j) = \alpha(i_{j-1}) = i_j$, so α' fixes i_j . Conversely, if α' moves k , then $k \notin Y$, so $k \in Y'$; but σ fixes every element of Y' . Hence $\alpha = \alpha'\sigma$ is a factorization into disjoint permutations. The number of points moved by α' is $k - r < k$, so the inductive hypothesis gives $\alpha' = \beta_1 \cdots \beta_t$ with $\beta_1, \dots, \beta_t$ disjoint cycles. Therefore $\alpha = \beta_1 \cdots \beta_t\sigma$ is a product of disjoint cycles. ■

Definition 1.1.17 (Complete factorization). A **complete factorization** of a permutation α is a factorization of α into disjoint cycles that contains one 1-cycle (i) for every i fixed by α .

The algorithm of Example 1.1.10 always yields a complete factorization. In a complete factorization $\alpha = \beta_1 \cdots \beta_t$, every symbol i between 1 and n occurs in exactly one β_λ .

Example 1.1.18. If $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 4 & 2 & 5 \end{pmatrix}$, the algorithm gives the complete factorization $\alpha = (1)(2\ 3\ 4)(5)$. By contrast the factorizations

$$\alpha = (2\ 3\ 4) = (1)(2\ 3\ 4) = (2\ 3\ 4)(5)$$

are *not* complete.

Write $\beta = (i_0\ i_1\ \dots\ i_{r-1})$. Then $i_1 = \beta(i_0)$, $i_2 = \beta(i_1) = \beta^2(i_0)$, and in general

$$i_k = \beta^k(i_0) \quad (k \leq r-1). \quad (1.1)$$

Since $\beta(i_{r-1}) = i_0$, equation (1.1) holds for all k if the subscripts j in i_j are read mod r .

Lemma 1.1.19 (Powers of a cycle)

Let $\beta = (i_0 \ i_1 \ \dots \ i_{r-1})$ be an r -cycle. Then $\beta^r = (1)$, and r is the smallest positive integer k with $\beta^k = (1)$.

Proof. Reading subscripts mod r , equation (1.1) gives $\beta^k(i_j) = i_{j+k}$ for all j and all $k \geq 1$. Taking $k = r$ yields $\beta^r(i_j) = i_j$ for every j ; since β fixes every other symbol, so does β^r , and hence $\beta^r = (1)$. If $0 < k < r$, then $\beta^k(i_0) = i_k \neq i_0$, because $i_0, \dots, i_{r-1}$ are distinct; therefore $\beta^k \neq (1)$. ■

Lemma 1.1.20 (i) Let $\alpha = \beta\delta$ be a factorization into disjoint permutations. If β moves i , then $\alpha^k(i) = \beta^k(i)$ for all $k \geq 1$.

(ii) If β and γ are cycles both of which move $i = i_0$, and if $\beta^k(i) = \gamma^k(i)$ for all $k \geq 1$, then $\beta = \gamma$.

Remark 1.1.21. The hypothesis in (ii) does *not* assume that β and γ have the same length; that is part of the conclusion.

Proof. (i) Since β moves i , disjointness implies δ fixes i , and hence so does every power of δ . Now β and δ commute by Lemma 1.1.13, so $(\beta\delta)^k(i) = \beta^k(\delta^k(i)) = \beta^k(i)$.

(ii) By (1.1), if $\beta = (i_0 \ i_1 \ \dots \ i_{r-1})$, then $i_k = \beta^k(i_0)$, and if $\gamma = (i_0 \ j_1 \ \dots \ j_{s-1})$, then $j_k = \gamma^k(i_0)$. We may assume $r \leq s$, so that $i_1 = j_1, \dots, i_{r-1} = j_{r-1}$. Since $j_r = \gamma^r(i_0) = \beta^r(i_0) = i_0$, it follows that $s = r$ and $j_k = i_k$ for all k . Therefore $\beta = \gamma$. ■

Theorem 1.1.22 (Uniqueness of the complete factorization)

Let $\alpha \in S_n$ and let $\alpha = \beta_1 \cdots \beta_t$ be a complete factorization into disjoint cycles. This factorization is unique except for the order in which the cycles occur.

Intuition . This is the analog, for permutations, of the fundamental theorem of arithmetic: cycles play the role of primes.

Proof. Let $\alpha = \gamma_1 \cdots \gamma_s$ be a second complete factorization into disjoint cycles. Every complete factorization has exactly one 1-cycle for each i fixed by α , so it suffices to prove, by induction on ℓ , the larger of t and s , that the cycles of length > 1 are uniquely determined by α .

The base step $\ell = 1$ is the hypothesis $\beta_1 = \alpha = \gamma_1$.

For the inductive step, if β_t moves $i = i_0$, then $\beta_t^k(i_0) = \alpha^k(i_0)$ for all $k \geq 1$, by Lemma 1.1.20(i). Some γ_j must also move i_0 ; since disjoint cycles commute, we may re-index so that γ_s moves i_0 , and then $\gamma_s^k(i_0) = \alpha^k(i_0)$ for all k . Now Lemma 1.1.20(ii) gives $\beta_t = \gamma_s$, so that cancellation (Proposition 1.1.6) yields

$$\beta_1 \cdots \beta_{t-1} = \gamma_1 \cdots \gamma_{s-1}.$$

By the inductive hypothesis $s = t$ and the γ 's may be re-indexed so that $\gamma_\lambda = \beta_\lambda$ for all λ . ■

Proposition 1.1.23 (Inverses) (i) $(i_1 \ i_2 \ \dots \ i_r)^{-1} = (i_r \ i_{r-1} \ \dots \ i_1)$.

(ii) If $\gamma \in S_n$ and $\gamma = \beta_1 \cdots \beta_k$, then $\gamma^{-1} = \beta_k^{-1} \cdots \beta_1^{-1}$ (note that the order of the factors is reversed).

Intuition . In the picture of a cycle as a clockwise rotation of a circle (Figure 1.1), the inverse is just the counterclockwise rotation.

Proof. (i) The composite $(i_1 i_2 \dots i_r)(i_r i_{r-1} \dots i_1)$ fixes every integer between 1 and n other than $i_1, \dots, i_r$; it sends $i_1 \mapsto i_r \mapsto i_1$, and for $j \geq 2$ it sends $i_j \mapsto i_{j-1} \mapsto i_j$. Thus every integer is fixed, so the composite is (1) . A similar argument treats the other order.

(ii) Induct on $k \geq 2$. For $k = 2$,

$$(\beta_1 \beta_2)(\beta_2^{-1} \beta_1^{-1}) = \beta_1(\beta_2 \beta_2^{-1})\beta_1^{-1} = \beta_1 \beta_1^{-1} = (1),$$

and similarly $(\beta_2^{-1} \beta_1^{-1})(\beta_1 \beta_2) = (1)$. For the inductive step, let $\delta = \beta_1 \cdots \beta_k$; then $\beta_1 \cdots \beta_k \beta_{k+1} = \delta \beta_{k+1}$, and so

$$(\beta_1 \cdots \beta_{k+1})^{-1} = (\delta \beta_{k+1})^{-1} = \beta_{k+1}^{-1} \delta^{-1} = \beta_{k+1}^{-1} (\beta_1 \cdots \beta_k)^{-1} = \beta_{k+1}^{-1} \beta_k^{-1} \cdots \beta_1^{-1}. \quad \blacksquare$$

Thus $(1 \ 2 \ 3 \ 4)^{-1} = (4 \ 3 \ 2 \ 1) = (1 \ 4 \ 3 \ 2)$, and $(1 \ 2)^{-1} = (2 \ 1) = (1 \ 2)$: every transposition is its own inverse.

Example 1.1.24. Proposition 1.1.23 applies in particular when the factors are disjoint cycles, in which case reversing the order is unnecessary because disjoint cycles commute. Thus if $\alpha = \left(\begin{smallmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 6 & 4 & 7 & 2 & 5 & 1 & 8 & 9 & 3 \end{smallmatrix}\right) = (1 \ 6)(2 \ 4)(3 \ 7 \ 8 \ 9)(5)$, then

$$\alpha^{-1} = (5)(9 \ 8 \ 7 \ 3)(4 \ 2)(6 \ 1) = (1 \ 6)(2 \ 4)(3 \ 9 \ 8 \ 7).$$

§1.1.ii. Cycle Structure and Conjugation

Definition 1.1.25 (Cycle structure). Two permutations $\alpha, \beta \in S_n$ have the same **cycle structure** if their complete factorizations have the same number of r -cycles for every $r \geq 1$.

There are $\frac{1}{r} [n(n-1) \cdots (n-r+1)]$ r -cycles in S_n (Rotman, Exercise 2.21). This formula counts permutations of a given cycle structure, provided one is careful when several cycles have the same length. For example, the number of permutations in S_4 of cycle structure $(a \ b)(c \ d)$ is

$$\frac{1}{2} \left[\frac{1}{2} (4 \times 3) \right] \cdot \left[\frac{1}{2} (2 \times 1) \right] = 3,$$

the extra factor $\frac{1}{2}$ occurring so that we do not count $(a \ b)(c \ d) = (c \ d)(a \ b)$ twice. Similarly, the number of permutations in S_n of the form $(a \ b)(c \ d)(e \ f)$ is

$$\frac{1}{3!2^3} [n(n-1)(n-2)(n-3)(n-4)(n-5)].$$

Cycle structure	Number	Cycle structure	Number
(1)	1	(1)	1
(1 2)	6	(1 2)	10
(1 2 3)	8	(1 2 3)	20
(1 2 3 4)	6	(1 2 3 4)	30
(1 2)(3 4)	3	(1 2 3 4 5)	24
	24	(1 2)(3 4 5)	20
		(1 2)(3 4)	15
			120
(a) Permutations in S_4 .		(b) Permutations in S_5 .	

Table 1.1.: Counting permutations by cycle structure.

Lemma 1.1.26

Let $\alpha, \gamma \in S_n$. For all i , if $\gamma: i \mapsto j$, then $\alpha\gamma\alpha^{-1}: \alpha(i) \mapsto \alpha(j)$.

Proof. $\alpha\gamma\alpha^{-1}(\alpha(i)) = \alpha\gamma(i) = \alpha(j)$. ■

Proposition 1.1.27

If $\gamma, \alpha \in S_n$, then $\alpha\gamma\alpha^{-1}$ has the same cycle structure as γ . In more detail, if the complete factorization of γ is

$$\gamma = \beta_1\beta_2 \cdots (i \ j \ \dots) \cdots \beta_t,$$

then $\alpha\gamma\alpha^{-1}$ is the permutation obtained from γ by applying α to every symbol occurring in the cycles of γ .

Remark 1.1.28. For example, if $\gamma = (1 \ 3)(2 \ 4 \ 7)(5)(6)$ and $\alpha = (2 \ 5 \ 6)(1 \ 4 \ 3)$, then

$$\alpha\gamma\alpha^{-1} = (\alpha 1 \ \alpha 3)(\alpha 2 \ \alpha 4 \ \alpha 7)(\alpha 5)(\alpha 6) = (4 \ 1)(5 \ 3 \ 7)(6)(2).$$

Proof. Let σ denote the permutation obtained from γ by applying α to all the symbols. If γ fixes i , then [Lemma 1.1.26](#) shows that $\alpha\gamma\alpha^{-1}$ fixes $\alpha(i)$, as does σ . Assume that γ moves i , say $\gamma(i) = j$, so that one of the cycles of γ is $(i \ j \ \dots)$; by definition, one of the cycles of σ is $(\alpha(i) \ \alpha(j) \ \dots)$, that is, $\sigma: \alpha(i) \mapsto \alpha(j)$. But [Lemma 1.1.26](#) says $\alpha\gamma\alpha^{-1}: \alpha(i) \mapsto \alpha(j)$, so σ and $\alpha\gamma\alpha^{-1}$ agree on every number of the form $\alpha(i)$. Since α is surjective, every $k \in X$ has this form, and so $\sigma = \alpha\gamma\alpha^{-1}$. ■

Proposition 1.1.29

If $\gamma, \gamma' \in S_n$, then γ and γ' have the same cycle structure if and only if there exists $\alpha \in S_n$ with $\gamma' = \alpha\gamma\alpha^{-1}$.

Proof. Sufficiency is [Proposition 1.1.27](#).

Conversely, assume γ and γ' have the same cycle structure; that is, $\gamma = \beta_1 \cdots \beta_t$ and $\gamma' = \sigma_1 \cdots \sigma_t$ are complete factorizations with β_λ and σ_λ of the same length for all $\lambda \leq t$. Write $\beta_\lambda = (i_1^\lambda, \dots, i_{r(\lambda)}^\lambda)$ and $\sigma_\lambda = (j_1^\lambda, \dots, j_{r(\lambda)}^\lambda)$, and define

$$\alpha(i_1^\lambda) = j_1^\lambda, \quad \alpha(i_2^\lambda) = j_2^\lambda, \quad \dots, \quad \alpha(i_{r(\lambda)}^\lambda) = j_{r(\lambda)}^\lambda$$

for all λ . Since $\beta_1 \cdots \beta_t$ is a *complete* factorization, every $i \in X = \{1, \dots, n\}$ occurs in exactly one β_λ ; hence $\alpha(i)$ is defined for every $i \in X$ and $\alpha: X \rightarrow X$ is a single-valued function. Since every $j \in X$ occurs in some σ_λ , α is surjective, hence a bijection, and so $\alpha \in S_n$. By [Proposition 1.1.27](#), the λ th cycle of $\alpha\gamma\alpha^{-1}$ is $(\alpha(i_1^\lambda) \ \dots \ \alpha(i_{r(\lambda)}^\lambda)) = \sigma_\lambda$. Therefore $\alpha\gamma\alpha^{-1} = \gamma'$. ■

Example 1.1.30. If $\gamma = (1 \ 2 \ 3)(4 \ 5)(6)$ and $\gamma' = (2 \ 5 \ 6)(3 \ 1)(4)$, then $\gamma' = \alpha\gamma\alpha^{-1}$ where

$$\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 5 & 6 & 3 & 1 & 4 \end{pmatrix} = (1 \ 2 \ 5)(3 \ 6 \ 4).$$

Note that there are other choices for α as well.

§1.1.iii. Parity

Proposition 1.1.31

If $n \geq 2$, then every $\alpha \in S_n$ is a product of transpositions.

Proof. Of course $(1) = (1\ 2)(1\ 2)$ is such a product, as is every transposition: $(i\ j) = (i\ j)(1\ 2)(1\ 2)$. By [Proposition 1.1.16](#) it suffices to factor an r -cycle β . If $r = 1$, then $\beta = (1\ 2)(1\ 2)$. If $r \geq 2$, then

$$\beta = (1\ 2\ \dots\ r) = (1\ r)(1\ r-1) \cdots (1\ 3)(1\ 2).$$

One checks this by evaluating each side: the left side sends $1 \mapsto 2$, and each of $(1\ r), (1\ r-1), \dots, (1\ 3)$ fixes 2, so the right side also sends $1 \mapsto 2$; and so on. ■

Remark 1.1.32 (Transposition factorizations are far from unique). Such a factorization is not as nice as the factorization into disjoint cycles. The transpositions occurring need not commute: $(1\ 2\ 3) = (1\ 3)(1\ 2) \neq (1\ 2)(1\ 3)$; and neither the factors nor their number is determined. For example, in S_4 ,

$$\begin{aligned} (1\ 2\ 3) &= (1\ 3)(1\ 2) \\ &= (2\ 3)(1\ 3) \\ &= (1\ 3)(4\ 2)(1\ 2)(1\ 4) \\ &= (1\ 3)(4\ 2)(1\ 2)(1\ 4)(2\ 3)(2\ 3). \end{aligned}$$

The only invariant that survives is the *parity* of the number of factors, and proving this is the goal of the rest of the section.

Example 1.1.33 (The 15-puzzle). The [15-puzzle](#) consists of a starting position, which is a 4×4 array of the numbers 1 through 15 together with a symbol $\#$ (a blank), and [simple moves](#). Consider the starting position

3	15	4	12
10	11	1	8
2	5	13	9
6	7	14	#

A simple move interchanges the blank with an adjacent symbol; here there are two beginning moves, interchanging $\#$ with 14 or with 9. One *wins* if, after a sequence of simple moves, the starting position is transformed into the standard array $1, 2, \dots, 15, \#$.

To analyze the game, note that the given array is really a permutation $\alpha \in S_{16}$ of $\{1, 2, \dots, 15, \#\}$: if the *squares* are labelled 1 through 15, $\#$, then $\alpha(i)$ is the symbol occupying the i th square. The position above is

$$\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 & 15 & \# \\ 3 & 15 & 4 & 12 & 10 & 11 & 1 & 8 & 2 & 5 & 13 & 9 & 6 & 7 & 14 & \# \end{pmatrix}.$$

Each simple move is a transposition τ that moves $\#$, and performing it from the position β yields the position $\tau\beta$. Hence to win one needs special transpositions $\tau_1, \dots, \tau_m$ with

$$\tau_m \cdots \tau_2 \tau_1 \alpha = (1).$$

Some starting positions can be won and others cannot, as we shall see in [Example 1.1.41](#).

Lemma 1.1.34

If $k, \ell \geq 0$ and the letters a, b, c_i, d_j are all distinct, then

$$(a \ b)(a \ c_1 \dots c_k \ b \ d_1 \dots d_\ell) = (a \ c_1 \dots c_k)(b \ d_1 \dots d_\ell)$$

and

$$(a \ b)(a \ c_1 \dots c_k)(b \ d_1 \dots d_\ell) = (a \ c_1 \dots c_k \ b \ d_1 \dots d_\ell).$$

Proof. The left side of the first equation sends

$$a \mapsto c_1 \mapsto c_1; \quad c_i \mapsto c_{i+1} \mapsto c_{i+1} \ (i < k); \quad c_k \mapsto b \mapsto a;$$

$$b \mapsto d_1 \mapsto d_1; \quad d_j \mapsto d_{j+1} \mapsto d_{j+1} \ (j < \ell); \quad d_\ell \mapsto a \mapsto b.$$

Evaluating the right side gives the same values on a, b , and all c_i, d_j , and each side fixes every other number, so the two sides are equal.

For the second equation, reverse the first and multiply both sides on the left by $(a \ b)$, using $(a \ b)(a \ b) = (1)$. ■

An illustration: $(1 \ 2)(1 \ 3 \ 4 \ 2 \ 5 \ 6 \ 7) = (1 \ 3 \ 4)(2 \ 5 \ 6 \ 7)$.

Definition 1.1.35 (Signum). If $\alpha \in S_n$ and $\alpha = \beta_1 \cdots \beta_t$ is a *complete* factorization into disjoint cycles, then the **signum** of α is

$$\text{sgn}(\alpha) = (-1)^{n-t}.$$

Theorem 1.1.22 shows that sgn is a well-defined function, for t is determined by α . If ε is a 1-cycle, then $t = n$ and $\text{sgn}(\varepsilon) = 1$. If τ is a transposition, then it moves two numbers and fixes the other $n - 2$, so $t = 1 + (n - 2) = n - 1$ and $\text{sgn}(\tau) = (-1)^{n-(n-1)} = -1$.

Lemma 1.1.36

If $\alpha, \tau \in S_n$ with τ a transposition, then $\text{sgn}(\tau\alpha) = -\text{sgn}(\alpha)$.

Proof. Let $\alpha = \beta_1 \cdots \beta_t$ be a complete factorization and let $\tau = (a \ b)$.

If a and b occur in the same β , say β_1 , then $\beta_1 = (a \ c_1 \dots c_k \ b \ d_1 \dots d_\ell)$ with $k, \ell \geq 0$, and **Lemma 1.1.34** gives $\tau\beta_1 = (a \ c_1 \dots c_k)(b \ d_1 \dots d_\ell)$. This exhibits a complete factorization of $\tau\alpha = (\tau\beta_1)\beta_2 \cdots \beta_t$ with $t + 1$ cycles, for β_1 has split in two. Therefore $\text{sgn}(\tau\alpha) = (-1)^{n-(t+1)} = -\text{sgn}(\alpha)$.

Otherwise a and b occur in different cycles, say $\beta_1 = (a \ c_1 \dots c_k)$ and $\beta_2 = (b \ d_1 \dots d_\ell)$. Then $\tau\alpha = (\tau\beta_1\beta_2)\beta_3 \cdots \beta_t$ and **Lemma 1.1.34** gives $\tau\beta_1\beta_2 = (a \ c_1 \dots c_k \ b \ d_1 \dots d_\ell)$. So $\tau\alpha$ has a complete factorization with $t - 1$ cycles, and $\text{sgn}(\tau\alpha) = (-1)^{n-(t-1)} = -\text{sgn}(\alpha)$. ■

Theorem 1.1.37 (sgn is multiplicative)

For all $\alpha, \beta \in S_n$,

$$\text{sgn}(\alpha\beta) = \text{sgn}(\alpha) \text{sgn}(\beta).$$

Proof. By **Proposition 1.1.31** we may write $\alpha = \tau_1 \cdots \tau_m$ as a product of transpositions; induct on m . The base step $m = 1$ is **Lemma 1.1.36**. If $m > 1$, the inductive hypothesis applies to

$\tau_2 \cdots \tau_m$, and

$$\begin{aligned}
 \operatorname{sgn}(\alpha\beta) &= \operatorname{sgn}(\tau_1 \cdots \tau_m \beta) \\
 &= -\operatorname{sgn}(\tau_2 \cdots \tau_m \beta) && \text{(Lemma 1.1.36)} \\
 &= -\operatorname{sgn}(\tau_2 \cdots \tau_m) \operatorname{sgn}(\beta) && \text{(induction)} \\
 &= \operatorname{sgn}(\tau_1 \cdots \tau_m) \operatorname{sgn}(\beta) && \text{(Lemma 1.1.36)} \\
 &= \operatorname{sgn}(\alpha) \operatorname{sgn}(\beta).
 \end{aligned}$$

It follows by induction on $k \geq 2$ that $\operatorname{sgn}(\alpha_1 \alpha_2 \cdots \alpha_k) = \operatorname{sgn}(\alpha_1) \operatorname{sgn}(\alpha_2) \cdots \operatorname{sgn}(\alpha_k)$.

Definition 1.1.38 (Even, odd, parity). A permutation $\alpha \in S_n$ is **even** if $\operatorname{sgn}(\alpha) = 1$, and **odd** if $\operatorname{sgn}(\alpha) = -1$. We say α and β have the same **parity** if both are even or both are odd.

Theorem 1.1.39 (i) Let $\alpha \in S_n$. If α is even, then α is a product of an even number of transpositions; if α is odd, then α is a product of an odd number of transpositions.
(ii) If $\alpha = \tau_1 \cdots \tau_q = \tau'_1 \cdots \tau'_p$ are two factorizations into transpositions, then q and p have the same parity.

Proof. (i) If $\alpha = \tau_1 \cdots \tau_q$, then **Theorem 1.1.37** gives $\operatorname{sgn}(\alpha) = \operatorname{sgn}(\tau_1) \cdots \operatorname{sgn}(\tau_q) = (-1)^q$, since every transposition is odd. Hence $\operatorname{sgn}(\alpha) = 1$ forces q even, and $\operatorname{sgn}(\alpha) = -1$ forces q odd.

(ii) If there were two factorizations, one into an odd and one into an even number of transpositions, then $\operatorname{sgn}(\alpha)$ would have two different values. ■

Corollary 1.1.40

Let $\alpha, \beta \in S_n$. If α and β have the same parity, then $\alpha\beta$ is even; if they have distinct parity, then $\alpha\beta$ is odd.

Proof. If $\operatorname{sgn}(\alpha) = (-1)^q$ and $\operatorname{sgn}(\beta) = (-1)^p$, then **Theorem 1.1.37** gives $\operatorname{sgn}(\alpha\beta) = (-1)^{q+p}$. ■

Example 1.1.41 (The 15-puzzle, resolved). One can show that if $\alpha \in S_{16}$ is the starting position of the 15-puzzle, then the game can be won if and only if α is an *even* permutation fixing #. One direction is clear. The blank # starts in position 16; each simple move takes # up, down, left, or right, so the total number of moves is $m = u + d + l + r$. If # is to return home, then $u = d$ and $l = r$, so $m = 2u + 2r$ is even. That is, if $\tau_m \cdots \tau_1 \alpha = (1)$, then m is even; hence $\alpha = \tau_1 \cdots \tau_m$ (because $\tau^{-1} = \tau$ for every transposition), and so α is even.

For the starting position of **Example 1.1.33**, the cycle notation is

$$\alpha = (1\ 3\ 4\ 12\ 9\ 2\ 15\ 14\ 7)(5\ 10)(6\ 11\ 13)(8)(\#),$$

so $t = 5$ and $\operatorname{sgn}(\alpha) = (-1)^{16-5} = -1$. Therefore α is odd, and the game starting with α *cannot* be won.

§1.2. Groups

Generalizations of the quadratic formula for cubics and quartics were discovered in the early 1500s. Over the next three centuries many tried to find analogous formulas for higher degree, but in 1824 N. H. Abel (1802–1829) proved that no such formula exists for the general polynomial of degree 5. In 1831 E. Galois (1811–1832) solved the problem completely, determining precisely

which polynomials admit such a formula; his fundamental idea was the invention of the notion of *group*. Since Galois's time groups have arisen throughout mathematics, for they are also the way to describe *symmetry*.

Intuition . The essence of a “product” is that two things are combined to form a third thing of the same kind: multiplication, addition and subtraction combine two numbers to give a number; composition combines two permutations to give a permutation.

Definition 1.2.1 (Binary operation). A **(binary) operation** on a set G is a function $*$: $G \times G \rightarrow G$.

In more detail, an operation assigns an element $*(x, y) \in G$ to each *ordered* pair (x, y) of elements of G ; it is more natural to write $x * y$. Thus composition of functions is the operation $(f, g) \mapsto g \circ f$, while multiplication, addition and subtraction are $(x, y) \mapsto xy$, $(x, y) \mapsto x + y$ and $(x, y) \mapsto x - y$. Composition and subtraction show why we want *ordered* pairs, for $x * y$ and $y * x$ may differ.

Note 1.2.2 (Law of substitution). An operation is, as any function, single-valued. Stated explicitly, this is called the **law of substitution**: if $x = x'$ and $y = y'$, then $x * y = x' * y'$.

Definition 1.2.3 (Group). A **group** is a set G equipped with an operation $*$ and a special element $e \in G$, called the **identity**, such that

- (i) the **associative law** holds: $a * (b * c) = (a * b) * c$ for all $a, b, c \in G$;
- (ii) $e * a = a$ for all $a \in G$;
- (iii) for every $a \in G$ there is $a' \in G$ with $a' * a = e$.

Remark 1.2.4. Only a *left* identity and *left* inverses are assumed. The two-sided statements $a * e = a$ and $a * a' = e$ are theorems, proved in [Proposition 1.2.8](#).

The set S_X of all permutations of a set X , with composition as operation and 1_X as identity, is a group: the symmetric group on X .

Intuition (Why abstraction pays). We are now at the precise point where algebra becomes *abstract* algebra. In contrast to the concrete group S_n , we will prove general results about groups without specifying either their elements or their operation. Theorems then apply to *many* different groups at once, and it is often simpler to work abstractly even when dealing with one particular concrete group (see [Example 1.2.19](#)).

Definition 1.2.5 (Abelian group). A group G is **abelian** if it satisfies the **commutative law**: $x * y = y * x$ for all $x, y \in G$.

The name honours Abel, who proved in 1827 a theorem equivalent to the existence of a formula for the roots of a polynomial whose Galois group is commutative. The groups S_n for $n \geq 3$ are *not* abelian, because $(1\ 2)(1\ 3) = (1\ 3\ 2)$ while $(1\ 3)(1\ 2) = (1\ 2\ 3)$.

§1.2.i. First Consequences of the Axioms

Associativity says $a * (b * c) = (a * b) * c$, so it is unambiguous to write $a * b * c$. Not all operations are associative: if $c \neq 0$, then $a - (b - c) \neq (a - b) - c$. The next lemma shows that some associativity carries over to products of four factors.

Lemma 1.2.6

If $*$ is an associative operation on a set G , then

$$(a * b) * (c * d) = [a * (b * c)] * d$$

for all $a, b, c, d \in G$.

Proof. Write $g = a * b$. Then $(a * b) * (c * d) = g * (c * d) = (g * c) * d = [(a * b) * c] * d = [a * (b * c)] * d$. ■

Lemma 1.2.7

If G is a group and $a \in G$ satisfies $a * a = a$, then $a = e$.

Proof. There is $a' \in G$ with $a' * a = e$. Multiplying both sides of $a * a = a$ on the left by a' gives $a' * (a * a) = a' * a = e$; but $a' * (a * a) = (a' * a) * a = e * a = a$. Hence $a = e$. ■

Proposition 1.2.8

Let G be a group with operation $*$ and identity e .

- (i) $a * a' = e$ for all $a \in G$.
- (ii) $a * e = a$ for all $a \in G$.
- (iii) If $e' \in G$ satisfies $e' * a = a$ for all $a \in G$, then $e' = e$; that is, the identity is unique.
- (iv) Let $a \in G$. If $b \in G$ satisfies $b * a = e$, then $b = a'$; that is, inverses are unique.

Proof. (i) By [Lemma 1.2.6](#),

$$(a * a') * (a * a') = [a * (a' * a)] * a' = (a * e) * a' = a * (e * a') = a * a',$$

and [Lemma 1.2.7](#) gives $a * a' = e$.

- (ii) Using part (i), $a * e = a * (a' * a) = (a * a') * a = e * a = a$.
- (iii) If $e' * a = a$ for all $a \in G$, then in particular $e' * e' = e'$, so $e' = e$ by [Lemma 1.2.7](#).
- (iv) By part (i), $a * a' = e$. Hence $b = b * e = b * (a * a') = (b * a) * a' = e * a' = a'$. ■

Definition 1.2.9 (Inverse). If G is a group and $a \in G$, then the unique element $a' \in G$ with $a' * a = e$ is called the **inverse** of a , denoted by a^{-1} .

Lemma 1.2.10

Let G be a group.

- (i) The **cancellation laws** hold: if $a, b, x \in G$ and either $x * a = x * b$ or $a * x = b * x$, then $a = b$.
- (ii) $(a^{-1})^{-1} = a$ for all $a \in G$.
- (iii) If $a, b \in G$, then $(a * b)^{-1} = b^{-1} * a^{-1}$. More generally, for all $n \geq 2$,

$$(a_1 * a_2 * \cdots * a_n)^{-1} = a_n^{-1} * \cdots * a_2^{-1} * a_1^{-1}.$$

Proof. (i)

$$a = e * a = (x^{-1} * x) * a = x^{-1} * (x * a) = x^{-1} * (x * b) = (x^{-1} * x) * b = e * b = b.$$

A similar proof, using $x * x^{-1} = e$, works when x is on the right.

(ii) By Proposition 1.2.8(i), $a * a^{-1} = e$. Uniqueness of inverses, Proposition 1.2.8(iv), says that $(a^{-1})^{-1}$ is the unique $x \in G$ with $x * a^{-1} = e$; therefore $(a^{-1})^{-1} = a$.

(iii) By Lemma 1.2.6,

$$(a * b) * (b^{-1} * a^{-1}) = [a * (b * b^{-1})] * a^{-1} = (a * e) * a^{-1} = a * a^{-1} = e,$$

so $(a * b)^{-1} = b^{-1} * a^{-1}$ by Proposition 1.2.8(iv). The general statement follows by induction on n . ■

Notation . From now on we usually write ab for $a * b$ and 1 for the identity. When a group is abelian we often use *additive* notation instead: an **additive group** is a set G with an operation $+$ and an element $0 \in G$ such that $a + (b + c) = (a + b) + c$, $0 + a = a$, and for every a there is $-a \in G$ with $(-a) + a = 0$. The inverse of a is written $-a$ rather than a^{-1} .

§1.2.ii. Examples of Groups

Example 1.2.11 (A first list of groups). (i) S_X , the set of all permutations of a set X , is a group under composition; in particular S_n is a group.

(ii) $\mathbb{Z}$ is an additive abelian group with $a * b = a + b$, identity $e = 0$, and inverse $-n$ of n . Similarly $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ are additive abelian groups.

(iii) $\mathbb{Q}^\times$, the set of all nonzero rationals, is an abelian group under ordinary multiplication, with identity 1 and inverse $1/r$ of r . Similarly $\mathbb{R}^\times$ is a multiplicative abelian group. Note that $\mathbb{Z}^\times$ is *not* a group, for no element other than ± 1 has a multiplicative inverse in $\mathbb{Z}^\times$.

(iv) The nonzero complex numbers $\mathbb{C}^\times$ form an abelian group under multiplication. For inverses, if $z = a + ib$ with $a, b \in \mathbb{R}$, let $\bar{z} = a - ib$ be its complex conjugate; then $z\bar{z} = a^2 + b^2$, so $z \neq 0$ if and only if $z\bar{z} \neq 0$, and for $z \neq 0$,

$$z^{-1} = \bar{z}/z\bar{z} = (a/z\bar{z}) - (b/z\bar{z})i.$$

(v) The **circle group** is $S^1 = \{z \in \mathbb{C} : |z| = 1\}$ under multiplication of complex numbers; the identity is 1 , and the inverse of a number of modulus 1 is its complex conjugate. Although S^1 is abelian we still write it multiplicatively.

(vi) For a positive integer n , let $\Gamma_n = \{\zeta^k : 0 \leq k < n\}$ be the set of all n th roots of unity, where $\zeta = e^{2\pi i/n} = \cos(2\pi/n) + i \sin(2\pi/n)$. De Moivre's theorem shows Γ_n is an abelian group under multiplication, the inverse of a root of unity being its complex conjugate.

(vii) The plane $\mathbb{R} \times \mathbb{R}$ is an additive abelian group under vector addition: $(x, y) + (x', y') = (x + x', y + y')$. The identity is the origin $O = (0, 0)$ and the inverse of $\mathbf{v} = (x, y)$ is $-\mathbf{v} = (-x, -y)$.

(viii) The **parity group** $\mathbb{P}$ has two elements, the words “even” and “odd”, with

$$\text{even} + \text{even} = \text{even} = \text{odd} + \text{odd}, \quad \text{even} + \text{odd} = \text{odd} = \text{odd} + \text{even}.$$

- (ix) The **Boolean group** $\mathcal{B}(X)$ of a set X (named after the logician G. Boole, 1815–1864) is the family of all subsets of X with addition given by symmetric difference $A + B = (A - B) \cup (B - A)$. Symmetric difference is commutative, the identity is $\emptyset$, and every element is its own inverse, since $A + A = \emptyset$. Thus $\mathcal{B}(X)$ is abelian.

Example 1.2.12 (Matrix groups and affine maps). (i) A (2×2) real matrix is $A = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$ with $a, b, c, d \in \mathbb{R}$, and the product is

$$AB = \begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} w & y \\ x & z \end{pmatrix} = \begin{pmatrix} aw + cx & ay + cz \\ bw + dx & by + dz \end{pmatrix};$$

each entry of AB is a dot product of a row of A with a column of B . The **determinant** of A is $\det(A) = ad - bc$, and A is **nonsingular** if $\det(A) \neq 0$. Since $\det(AB) = \det(A)\det(B)$, a product of nonsingular matrices is nonsingular. The set $GL(2, \mathbb{R})$ of all nonsingular matrices, under matrix multiplication, is a *nonabelian* group, the 2×2 real **general linear group**: the identity is $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and

$$A^{-1} = \begin{pmatrix} d/\Delta & -c/\Delta \\ -b/\Delta & a/\Delta \end{pmatrix}, \quad \Delta = ad - bc = \det(A).$$

- (ii) Allowing entries in $\mathbb{Q}$ or $\mathbb{C}$ gives $GL(2, \mathbb{Q})$ and $GL(2, \mathbb{C})$; one may even allow entries in $\mathbb{Z}$, in which case $GL(2, \mathbb{Z})$ is defined to be the set of such matrices with determinant ± 1 (so that A^{-1} also has integer entries). More generally, all nonsingular $n \times n$ matrices form a group $GL(n, k)$.
- (iii) All **special orthogonal** matrices, that is, all matrices $A = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$, form the 2×2 **special orthogonal group** $SO(2, \mathbb{R})$. Indeed

$$\begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix} \begin{pmatrix} \cos \beta & -\sin \beta \\ \sin \beta & \cos \beta \end{pmatrix} = \begin{pmatrix} \cos(\alpha + \beta) & -\sin(\alpha + \beta) \\ \sin(\alpha + \beta) & \cos(\alpha + \beta) \end{pmatrix}$$

by the addition theorems for sine and cosine; this calculation also shows that $SO(2, \mathbb{R})$ is abelian. The identity matrix is special orthogonal, and so is the inverse of a special orthogonal matrix. We shall see that $SO(2, \mathbb{R})$ is a disguised version of the circle group S^1 , and that it consists of all rotations of the plane about the origin.

- (iv) The **affine group** $\text{Aff}(1, \mathbb{R})$ consists of all **affine maps** $f_{a,b}: \mathbb{R} \rightarrow \mathbb{R}$, $f_{a,b}(x) = ax + b$, with a, b fixed reals and $a \neq 0$. It is a group under composition: if $f_{c,d}(x) = cx + d$, then

$$f_{a,b}f_{c,d}(x) = a(cx + d) + b = acx + (ad + b) = f_{ac, ad+b}(x),$$

and $ac \neq 0$; the identity is $1_{\mathbb{R}} = f_{1,0}$, and the inverse of $f_{a,b}$ is $f_{a^{-1}, -a^{-1}b}$. The composition is reminiscent of matrix multiplication:

$$\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} c & d \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} ac & ad + b \\ 0 & 1 \end{pmatrix}.$$

§1.2.iii. Generalized Associativity and Powers

Definition 1.2.13 (*n-expression, ultimate product*). An **n-expression** is an n -tuple $(a_1, \dots, a_n)$ in the product of n copies of G . It yields elements of G as follows: choose two adjacent entries of the tuple, multiply them, and obtain an $(n - 1)$ -expression; repeat until a 2-expression (W, X) is reached, and then form $WX \in G$. Such an element WX is called an **ultimate product** derived from the expression. The expression **needs no parentheses** if all ultimate products derived from it are equal.

For example, from (a, b, c, d) one can reach $[(ab)c]d$, $(ab)(cd)$, $a[(bc)d]$, and so on. To say that an operation is associative is exactly to say that the two ultimate products from every 3-expression agree; it is not obvious that all ultimate products from a longer expression agree.

Theorem 1.2.14 (Generalized associativity)

If $n \geq 3$, then every n -expression $(a_1, \dots, a_n)$ in a group G needs no parentheses.

Remark 1.2.15. Neither the identity element nor inverses are used in the proof; the hypothesis can be weakened to G being a **semigroup**, that is, a nonempty set with an associative binary operation.

Proof. Induct on n (second form). The base step $n = 3$ is associativity. For the inductive step, consider two 2-expressions obtained from $(a_1, \dots, a_n)$:

$$(W, X) = (a_1 \cdots a_i, a_{i+1} \cdots a_n) \quad \text{and} \quad (Y, Z) = (a_1 \cdots a_j, a_{j+1} \cdots a_n).$$

We must prove $WX = YZ$. By induction each of W, X, Y, Z is the one and only ultimate product of an m -expression with $m < n$. Without loss of generality $i \leq j$. If $i = j$, then $W = Y$ and $X = Z$, so $WX = YZ$.

Assume $i < j$. Let A be the ultimate product of $(a_1, \dots, a_i)$, let B be that of $(a_{i+1}, \dots, a_j)$, and let C be that of $(a_{j+1}, \dots, a_n)$; these are unambiguous by the inductive hypothesis. Then $W = A$, $Z = C$, $X = BC$ and $Y = AB$, so $WX = A(BC)$ and $YZ = (AB)C$. The base step $n = 3$ gives $WX = YZ$. ■

Definition 1.2.16 (Powers). If G is a group and $a \in G$, define a^n for $n \geq 1$ inductively by $a^1 = a$ and $a^{n+1} = aa^n$; define $a^0 = 1$ and, for $n > 0$, $a^{-n} = (a^{-1})^n$.

One shows $(a^{-1})^n = (a^n)^{-1}$; this is a special case of [Lemma 1.2.10\(iii\)](#). There is a hidden complication: we defined $a^3 = aa^2 = a(aa)$, but $(aa)a = a^2a$ is another reasonable contender. Associativity makes them equal, and generalized associativity shows that all powers are unambiguously defined.

Corollary 1.2.17

If G is a group, $a \in G$, and $m, n \geq 1$, then

$$a^{m+n} = a^m a^n \quad \text{and} \quad (a^m)^n = a^{mn}.$$

Proof. Both a^{m+n} and $a^m a^n$ are ultimate products of the expression having $m+n$ factors each equal to a ; both $(a^m)^n$ and a^{mn} arise from the expression having mn factors each equal to a . Apply [Theorem 1.2.14](#). ■

In particular any two powers of the same element commute: $a^m a^n = a^{m+n} = a^{n+m} = a^n a^m$.

Proposition 1.2.18 (Laws of exponents)

Let G be a group, $a, b \in G$, and let m, n be (not necessarily positive) integers.

- (i) If a and b commute, then $(ab)^n = a^n b^n$.
- (ii) $(a^n)^m = a^{mn}$.
- (iii) $a^m a^n = a^{m+n}$.

Proof. Exercises for the reader. ■

Notation . If the operation is $+$, it is more natural to write na for $a + a + \cdots + a$. In additive notation [Proposition 1.2.18](#) reads

$$n(a + b) = na + nb, \quad m(na) = (mn)a, \quad ma + na = (m + n)a.$$

§1.2.iv. The Order of an Element

Example 1.2.19 (Shuffling a deck). Suppose a deck of cards is shuffled, changing the order from $1, 2, 3, 4, \dots, 52$ to $2, 1, 4, 3, \dots, 52, 51$. Shuffling again the same way restores the original order. In fact for *any* permutation α of the 52 cards, repeating α sufficiently often restores the deck. One proof uses our knowledge of permutations: write $\alpha = \beta_1 \cdots \beta_t$ with β_i a disjoint r_i -cycle. Then $\beta_i^{r_i} = (1)$ by [Lemma 1.1.19](#), so $\beta_i^k = (1)$ for $k = r_1 \cdots r_t$; since disjoint cycles commute, [Proposition 1.2.18\(i\)](#) gives

$$\alpha^k = (\beta_1 \cdots \beta_t)^k = \beta_1^k \cdots \beta_t^k = (1).$$

Here is a more general result with a *simpler* proof (abstract algebra can be easier than algebra): if G is a finite group and $a \in G$, then $a^k = 1$ for some $k \geq 1$. Indeed, consider $1, a, a^2, \dots$. Since G is finite there must be a repetition, say $a^m = a^n$ with $m > n$; then $1 = a^m a^{-n} = a^{m-n}$.

Definition 1.2.20 (Order of an element). Let G be a group and $a \in G$. If $a^k = 1$ for some $k \geq 1$, then the smallest such exponent $k \geq 1$ is called the **order** of a ; if no such power exists, a is said to have **infinite order**.

Every element of a finite group has finite order, by [Example 1.2.19](#). In any group the identity is the only element of order 1; an element has order 2 if and only if it is not the identity and equals its own inverse. The matrix $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \in GL(2, \mathbb{R})$ has infinite order, since $A^k = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix} \neq I$ for all $k \geq 1$.

Lemma 1.2.21

Let G be a group and let $a \in G$ have finite order k . If $a^n = 1$, then $k \mid n$. In fact $\{n \in \mathbb{Z} : a^n = 1\}$ is the set of all multiples of k .

Proof. Let $I = \{n \in \mathbb{Z} : a^n = 1\}$. Then $0 \in I$ because $a^0 = 1$; if $n, m \in I$, then $a^{n-m} = a^n a^{-m} = 1$, so $n - m \in I$; and if $n \in I$, $q \in \mathbb{Z}$, then $a^{qn} = (a^n)^q = 1$, so $qn \in I$. Therefore I consists of all the multiples of its smallest positive element, which by definition is the order k of a . Hence $a^n = 1$ implies $k \mid n$. ■

Proposition 1.2.22 (Order of a permutation)

Let $\alpha \in S_n$.

- (i) If α is an r -cycle, then α has order r .
- (ii) If $\alpha = \beta_1 \cdots \beta_t$ is a product of disjoint r_i -cycles β_i , then α has order $m = \text{lcm}(r_1, \dots, r_t)$.
- (iii) If p is a prime, then α has order p if and only if it is a p -cycle or a product of disjoint p -cycles.

Proof. (i) This is [Lemma 1.1.19](#).

(ii) Each β_i has order r_i by (i). Suppose $\alpha^M = (1)$. Since the β_i commute, $(1) = \alpha^M = \beta_1^M \cdots \beta_t^M$. The β_i are disjoint, hence so are their powers, and a product of disjoint permutations can be the identity only if every factor is; thus $\beta_i^M = (1)$ for each i , and then [Lemma 1.2.21](#) gives $r_i \mid M$ for all i , so M is a common multiple of $r_1, \dots, r_t$. Conversely, if $m = \text{lcm}(r_1, \dots, r_t)$, then $\alpha^m = (1)$. Hence α has order m .

(iii) Write α as a product of disjoint cycles and use (ii). ■

For example, a permutation in S_n has order 2 if and only if it is either a transposition or a product of disjoint transpositions. We can now augment [Table 1.1](#).

Cycle structure	Number	Order	Parity
(1)	1	1	Even
(1 2)	10	2	Odd
(1 2 3)	20	3	Even
(1 2 3 4)	30	4	Odd
(1 2 3 4 5)	24	5	Even
(1 2)(3 4 5)	20	6	Odd
(1 2)(3 4)	15	2	Even
120			

Table 1.2.: Permutations in S_5 .

§1.2.v. Symmetry

What do we mean when we say that an isosceles triangle $\triangle$ is symmetric? Put $\triangle = \triangle ABC$ with base AB on the x -axis and the y -axis the perpendicular bisector of AB . Close your eyes; let $\triangle$ be reflected in the y -axis (interchanging A and B); open your eyes. You cannot tell that $\triangle$ has been reflected: $\triangle$ is symmetric about the y -axis. Had $\triangle$ been reflected in the x -axis it would be obvious. Reflection is a special kind of *isometry*.

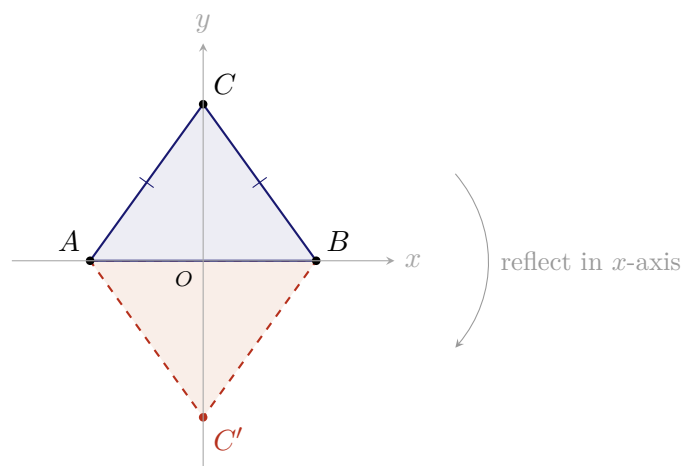


Figure 1.2.: The isosceles triangle $\triangle ABC$ is symmetric about the y -axis; its reflection in the x -axis (dashed) is visibly different.

Definition 1.2.23 (Isometry). An **isometry** of the plane is a function $\varphi: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that is *distance preserving*: for all $P = (a, b)$ and $Q = (c, d)$,

$$\|\varphi(P) - \varphi(Q)\| = \|P - Q\|, \quad \text{where } \|P - Q\| = \sqrt{(a - c)^2 + (b - d)^2}.$$

Write $P \cdot Q = ac + bd$ for the dot product of $P = (a, b)$ and $Q = (c, d)$. It is symmetric and bilinear, and $P \cdot P = a^2 + b^2 = \|P\|^2$, where $O = (0, 0)$ is the origin and $\|P\| = \|P - O\|$. Expanding,

$$\|P - Q\|^2 = (P - Q) \cdot (P - Q) = P \cdot P - 2(P \cdot Q) + Q \cdot Q. \quad (1.2)$$

Lemma 1.2.24 (Isometries fixing O preserve dot products)

Let $\varphi: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a function.

- (i) If φ is an isometry with $\varphi(O) = O$, then $\varphi(P) \cdot \varphi(Q) = P \cdot Q$ for all P, Q .
- (ii) Conversely, if $\varphi(P) \cdot \varphi(Q) = P \cdot Q$ for all P, Q , then φ is an isometry and $\varphi(O) = O$.

Proof. (i) Putting $Q = O$ in the definition of an isometry gives $\|\varphi(P)\| = \|\varphi(P) - \varphi(O)\| = \|P - O\| = \|P\|$, so $\varphi(P) \cdot \varphi(P) = P \cdot P$ for every P . Now expand both sides of $\|\varphi(P) - \varphi(Q)\|^2 = \|P - Q\|^2$ using (1.2):

$$\varphi(P) \cdot \varphi(P) - 2(\varphi(P) \cdot \varphi(Q)) + \varphi(Q) \cdot \varphi(Q) = P \cdot P - 2(P \cdot Q) + Q \cdot Q.$$

The first terms on the two sides are equal, and so are the last terms; cancelling them and dividing by -2 gives $\varphi(P) \cdot \varphi(Q) = P \cdot Q$.

(ii) By (1.2) and the hypothesis,

$$\begin{aligned} \|\varphi(P) - \varphi(Q)\|^2 &= \varphi(P) \cdot \varphi(P) - 2(\varphi(P) \cdot \varphi(Q)) + \varphi(Q) \cdot \varphi(Q) \\ &= P \cdot P - 2(P \cdot Q) + Q \cdot Q = \|P - Q\|^2, \end{aligned}$$

and taking nonnegative square roots shows that φ is an isometry. Moreover $\|\varphi(O)\|^2 = \varphi(O) \cdot \varphi(O) = O \cdot O = 0$, so $\varphi(O) = O$. ■

Remark 1.2.25. The hypothesis $\varphi(O) = O$ in (i) cannot be dropped. The map $\varphi(U) = U + V$ with $V = (1, 0)$ is an isometry, since $\|(P + V) - (Q + V)\| = \|P - Q\|$, but $\varphi(O) \cdot \varphi(O) = V \cdot V = 1 \neq 0 = O \cdot O$.

Dot products measure angles. Let $P, Q \neq O$, and write $P = \|P\|(\cos \alpha, \sin \alpha)$ and $Q = \|Q\|(\cos \beta, \sin \beta)$ in polar coordinates. Then

$$P \cdot Q = \|P\|\|Q\|(\cos \alpha \cos \beta + \sin \alpha \sin \beta) = \|P\|\|Q\|\cos(\alpha - \beta).$$

The angle $\theta \in [0, \pi]$ between P and Q (that is, $\angle POQ$) differs from $\pm(\alpha - \beta)$ by a multiple of 2π ; since $\cos$ is even and 2π -periodic, $\cos \theta = \cos(\alpha - \beta)$. Hence

$$P \cdot Q = \|P\|\|Q\|\cos \theta, \quad \text{that is,} \quad \cos \theta = \frac{P \cdot Q}{\|P\|\|Q\|}. \quad (1.3)$$

Since $\cos$ is injective on $[0, \pi]$, this determines θ . In particular, P and Q are orthogonal ($\theta = \pi/2$) if and only if $P \cdot Q = 0$.

Corollary 1.2.26 (Isometries preserve angles) (i) If φ is an isometry with $\varphi(O) = O$ and $P, Q \neq O$, then the angle between $\varphi(P)$ and $\varphi(Q)$ equals the angle between P and Q .

(ii) More generally, if φ is any isometry and P, Q, R are points with $Q, R \neq P$, then $\angle QPR = \angle \varphi(Q)\varphi(P)\varphi(R)$.

In particular, every isometry preserves perpendicularity.

Proof. (i) By Lemma 1.2.24, $\varphi(P) \cdot \varphi(Q) = P \cdot Q$, and, as in its proof, $\|\varphi(P)\| = \|P\| \neq 0$ and $\|\varphi(Q)\| = \|Q\| \neq 0$. If θ and θ' are the angles between P, Q and between $\varphi(P), \varphi(Q)$, then (1.3) gives

$$\cos \theta' = \frac{\varphi(P) \cdot \varphi(Q)}{\|\varphi(P)\| \|\varphi(Q)\|} = \frac{P \cdot Q}{\|P\| \|Q\|} = \cos \theta,$$

and $\theta' = \theta$ because both lie in $[0, \pi]$.

(ii) The angle $\angle QPR$ is the angle between the vectors $Q - P$ and $R - P$. Since $(Q - P) - (R - P) = Q - R$, applying (1.2) to $Q - P$ and $R - P$ gives

$$(Q - P) \cdot (R - P) = \frac{1}{2}(\|Q - P\|^2 + \|R - P\|^2 - \|Q - R\|^2).$$

The right side involves only distances, as do $\|Q - P\|$ and $\|R - P\|$, so none of them change when P, Q, R are replaced by $\varphi(P), \varphi(Q), \varphi(R)$. By (1.3), the two angles have the same cosine, hence are equal.

Finally, perpendicularity means meeting at an angle of $\pi/2$, which is preserved by (ii). ■

Notation . Denote the set of all isometries of the plane by $\text{Isom}(\mathbb{R}^2)$. Its subset of isometries φ with $\varphi(O) = O$ is called the **orthogonal group** of the plane, denoted $O_2(\mathbb{R})$. If $P \neq Q$ are points, let $L[P, Q]$ denote the line they determine and PQ the line segment with endpoints P and Q .

Example 1.2.27 (Rotations, reflections, translations). (i) Given an angle θ , the **rotation** R_θ about the origin O is defined by $R_\theta(O) = O$, and for $P \neq O$, by rotating the segment PO through θ (counterclockwise if $\theta > 0$). One can rotate about any point of the plane.

(ii) The **reflection** ρ_ℓ in a line ℓ , called its **axis**, fixes each point of ℓ ; if $P \notin \ell$, then $\rho_\ell(P) = P'$ where ℓ is the perpendicular bisector of PP' . Pretending ℓ is a mirror, P' is the mirror image of P . Now $\rho_\ell \in \text{Isom}(\mathbb{R}^2)$, and if ℓ passes through the origin, then $\rho_\ell \in O_2(\mathbb{R})$.

(iii) Given a point V , **translation** by V is $\tau_V: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $\tau_V(U) = U + V$. A translation fixes the origin if and only if $V = O$, so the identity is the only translation that is also a rotation.

Lemma 1.2.28

If φ is an isometry of the plane, then distinct points P, Q, R are collinear if and only if $\varphi(P), \varphi(Q), \varphi(R)$ are collinear.

Proof. Suppose P, Q, R are collinear, with R between P and Q , so that $\|P - Q\| = \|P - R\| + \|R - Q\|$. If $\varphi(P), \varphi(Q), \varphi(R)$ were not collinear, they would be the vertices of a triangle, and the triangle inequality would give

$$\|\varphi(P) - \varphi(Q)\| < \|\varphi(P) - \varphi(R)\| + \|\varphi(R) - \varphi(Q)\|,$$

contradicting that φ preserves distance.

We can prove the other direction with similar method. ■

Every isometry φ is injective: if $P \neq Q$, then $\|\varphi(P) - \varphi(Q)\| = \|P - Q\| \neq 0$. Surjectivity is less obvious.

Proposition 1.2.29

Every isometry φ of the plane fixing the origin is a linear transformation.

Proof. Let $C_d = \{Q \in \mathbb{R}^2 : \|Q - O\| = d\}$ be the circle of radius $d > 0$ centred at O . If $P \in C_d$, then $d = \|\varphi(P) - \varphi(O)\| = \|\varphi(P) - O\|$, so $\varphi(C_d) \subseteq C_d$.

Let $P \neq O$ and $r \in \mathbb{R}$. If $\|P - O\| = p$, then $\|rP - O\| = |r|p$, so $rP \in L[O, P] \cap C_{|r|p}$. Since φ preserves collinearity (Lemma 1.2.28), $\varphi(L[O, P] \cap C_{|r|p}) \subseteq L[O, \varphi(P)] \cap C_{|r|p}$, and a line meets a circle in at most two points; hence $\varphi(rP) = \pm r\varphi(P)$. To eliminate the minus sign, suppose $r > 0$. The origin lies between $-rP$ and P , so the distance from $-rP$ to P is $rp + p$, whereas the distance from rP to P is $|rp - p|$. Since $rp + p \neq |rp - p|$ and φ preserves distance, $\varphi(rP) \neq -r\varphi(P)$. A similar argument works for $r < 0$. Therefore $\varphi(rP) = r\varphi(P)$.

It remains to prove $\varphi(P + Q) = \varphi(P) + \varphi(Q)$. If O, P, Q are collinear, choose U on $L[O, P]$ at distance 1 from the origin, so $P = pU$, $Q = qU$ and $P + Q = (p + q)U$. Since φ preserves scalar multiplication,

$$\varphi(P) + \varphi(Q) = p\varphi(U) + q\varphi(U) = (p + q)\varphi(U) = \varphi(P + Q).$$

If O, P, Q are not collinear, then $P + Q$ is the point S such that O, P, Q, S are the vertices of a parallelogram (parallelogram law). Since φ preserves distance, $O = \varphi(O)$, $\varphi(P)$, $\varphi(Q)$, $\varphi(S)$ are also the vertices of a parallelogram, so $\varphi(S) = \varphi(P) + \varphi(Q)$; that is, $\varphi(P + Q) = \varphi(P) + \varphi(Q)$. ■

Corollary 1.2.30

Every isometry $\varphi: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a bijection, and every isometry fixing O is a nonsingular linear transformation.

Proof. First assume $\varphi(O) = O$, so φ is linear by Proposition 1.2.29. Since φ is injective, $P = \varphi(e_1)$, $Q = \varphi(e_2)$ is a basis of $\mathbb{R}^2$, where $e_1 = (1, 0)$, $e_2 = (0, 1)$. Hence $\psi: aP + bQ \mapsto ae_1 + be_2$ is a well-defined function inverse to φ ; so φ is a bijection and is nonsingular.

Now let φ be any isometry, say $\varphi(O) = U$. Then $\tau_{-U} \circ \varphi$ fixes O , so $\tau_{-U} \circ \varphi = \theta$ is a nonsingular linear transformation, and $\varphi = \tau_U \circ \theta$ is a composite of bijections. ■

Proposition 1.2.31

Both $\text{Isom}(\mathbb{R}^2)$ and $O_2(\mathbb{R})$ are groups under composition.

Proof. Clearly $1_{\mathbb{R}^2}$ is an isometry. If φ' and φ are isometries, then

$$\|(\varphi'\varphi)(P) - (\varphi'\varphi)(Q)\| = \|\varphi(P) - \varphi(Q)\| = \|P - Q\|,$$

so composition is an operation on $\text{Isom}(\mathbb{R}^2)$. Each φ is a bijection by Corollary 1.2.30, and φ^{-1} is also an isometry: $\|P - Q\| = \|\varphi(\varphi^{-1}(P)) - \varphi(\varphi^{-1}(Q))\| = \|\varphi^{-1}(P) - \varphi^{-1}(Q)\|$. Composition of functions is always associative. The same proof applies to $O_2(\mathbb{R})$. ■

Corollary 1.2.32 (i) If O, P, Q are noncollinear points and φ, ψ are isometries of the plane with $\varphi(O) = \psi(O) = O$, $\varphi(P) = \psi(P)$ and $\varphi(Q) = \psi(Q)$, then $\varphi = \psi$.

(ii) More generally, if A, B, C are noncollinear points and φ, ψ are isometries of the plane agreeing on A, B and C , then $\varphi = \psi$.

Proof. (i) Since φ and ψ fix O , both are linear transformations by [Proposition 1.2.29](#). Since O, P, Q are noncollinear, the list P, Q is linearly independent in $\mathbb{R}^2$: a nontrivial relation $aP + bQ = O$ would put P and Q on a common line through O . As $\dim(\mathbb{R}^2) = 2$, the list P, Q is a basis, and two linear transformations agreeing on a basis are equal.

(ii) Let $U = \varphi(A) = \psi(A)$, and set $\varphi' = \tau_{-U} \circ \varphi \circ \tau_A$ and $\psi' = \tau_{-U} \circ \psi \circ \tau_A$. These are composites of isometries, hence isometries, and $\varphi'(O) = \varphi(A) - U = O$; likewise $\psi'(O) = O$. Moreover

$$\varphi'(B - A) = \varphi(B) - U = \psi(B) - U = \psi'(B - A),$$

and similarly $\varphi'(C - A) = \psi'(C - A)$. The points $O, B - A, C - A$ are the translates of A, B, C by $-A$, so they are noncollinear. By (i), $\varphi' = \psi'$, and therefore $\varphi = \tau_U \circ \varphi' \circ \tau_{-A} = \tau_U \circ \psi' \circ \tau_{-A} = \psi$. ■

Remark 1.2.33. In (i) the hypothesis $\varphi(O) = \psi(O) = O$ cannot be dropped, and in (ii) two points are not enough. If O, P, Q are noncollinear, the identity and the reflection $\rho_{L[P,Q]}$ both fix P and Q , yet they are different: the reflection moves $O \notin L[P, Q]$.

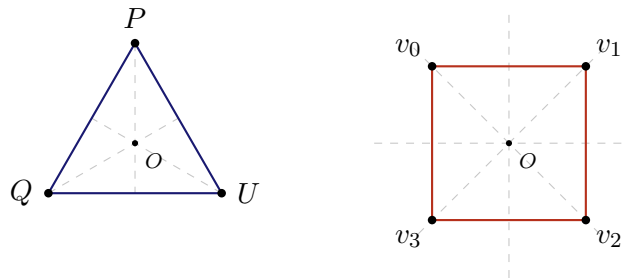
Example 1.2.34 (Triangles). If $\triangle$ is a triangle with vertices P, Q, U and φ is an isometry, then $\varphi(\triangle)$ is the triangle with vertices $\varphi(P), \varphi(Q), \varphi(U)$. If moreover $\varphi(\triangle) = \triangle$, then φ permutes the vertices. Assume the centre of $\triangle$ is O .

- If $\triangle$ is *isosceles* with equal sides PQ and PU , and ρ_ℓ is the reflection with axis $\ell = L[O, P]$, then $\rho_\ell(\triangle) = \triangle$; we may describe ρ_ℓ by the transposition $(Q\ U)$. On the other hand $\rho_{\ell'}(\triangle) \neq \triangle$ for $\ell' = L[O, Q]$.
- If $\triangle$ is *equilateral*, then also $\rho_{L[O,Q]}$ and $\rho_{L[O,U]}$ carry $\triangle$ to itself (described by $(P\ U)$ and $(P\ Q)$), and so do the rotations about O by 120° and 240° (described by $(P\ Q\ U)$ and $(P\ U\ Q)$).
- If $\triangle$ is *scalene*, then $\varphi(\triangle) = \triangle$ implies $\varphi = 1$.

So an equilateral triangle is “more symmetric” than an isosceles triangle, which in turn is “more symmetric” than a scalene one.

Definition 1.2.35 (Symmetry group). The **symmetry group** $\Sigma(\Omega)$ of a figure Ω in the plane is the set of all isometries φ of the plane with $\varphi(\Omega) = \Omega$. The elements of $\Sigma(\Omega)$ are called **symmetries** of Ω .

It is straightforward to see that $\Sigma(\Omega)$ is always a group.



$$\Sigma(\pi_3) = D_6, |\Sigma(\pi_3)| = 6 \quad \Sigma(\pi_4) = D_8, |\Sigma(\pi_4)| = 8$$

Figure 1.3.: Reflection axes of a regular 3-gon and a regular 4-gon.

Example 1.2.36 (Symmetry groups of regular polygons). (i) A regular 3-gon π_3 is an equilateral triangle, and $|\Sigma(\pi_3)| = 6$, as we saw in [Example 1.2.34](#).

- (ii) Let π_4 be a square with vertices $\{v_0, v_1, v_2, v_3\}$, drawn with centre at the origin and sides parallel to the axes. Every $\varphi \in \Sigma(\pi_4)$ permutes the vertices, and φ is determined by $\{\varphi(v_i)\}$, so there are at most $24 = 4!$ symmetries. Not every permutation arises, however: if v_i and v_j are adjacent, then $\|v_i - v_j\| = 1$, whereas $\|v_0 - v_2\| = \sqrt{2} = \|v_1 - v_3\|$; since isometries preserve distance, φ must preserve adjacency. There are exactly eight symmetries ([Theorem 1.2.39](#)): the identity, the three rotations about O by $90^\circ, 180^\circ, 270^\circ$, and four reflections with axes $L[v_0, v_2]$, $L[v_1, v_3]$, the x -axis and the y -axis. This group is the **dihedral group with 8 elements**, denoted D_8 .
- (iii) The symmetry group of a regular pentagon π_5 with vertices $v_0, \dots, v_4$ and centre O has 10 elements: the rotations about the origin by $(72j)^\circ$ for $0 \leq j \leq 4$, together with the reflections with axes $L[O, v_k]$ for $0 \leq k \leq 4$. It is denoted D_{10} .

Definition 1.2.37 (Dihedral group). A group D_{2n} with exactly $2n$ elements is called a **dihedral group** if it contains an element a of order n and an element b of order 2 such that

$$bab = a^{-1}.$$

Note 1.2.38. The definition is purely algebraic: it does not depend on geometry. If $n = 2$, then D_4 is abelian; if $n \geq 3$, then D_{2n} is not abelian. One can show that there is essentially only one dihedral group with $2n$ elements (any two such are isomorphic). The name comes from F. Klein's *dihedron*, two congruent regular polygons of zero thickness pasted together.

Theorem 1.2.39

The symmetry group $\Sigma(\pi_n)$ of a regular n -gon π_n is a dihedral group with $2n$ elements.

Proof. Let π_n have vertices $v_0, \dots, v_{n-1}$ and centre O . Define a to be the rotation about O by $(360/n)^\circ$,

$$a(v_i) = \begin{cases} v_{i+1} & \text{if } 0 \leq i < n-1, \\ v_0 & \text{if } i = n-1, \end{cases} \quad \text{and} \quad b(v_i) = \begin{cases} v_0 & \text{if } i = 0, \\ v_{n-i} & \text{if } 1 \leq i \leq n-1, \end{cases}$$

where b is the reflection with axis $L[O, v_0]$. Clearly a has order n and b has order 2. There are n distinct symmetries $1, a, a^2, \dots, a^{n-1}$, and $b, ab, a^2b, \dots, a^{n-1}b$ are also all distinct by cancellation. If $a^s = a^r b$ with $0 \leq r \leq n-1$ and $s = 0, 1$, then $a^s(v_i) = a^r b(v_i)$ for all i ; now $a^s(v_0) = v_s$ while $a^r b(v_0) = v_{r-1}$, forcing $s = r-1$; but with $i = 1$ we get $a^{r-1}(v_1) = v_{r+1}$ while $a^r b(v_1) = v_{r-1}$, a contradiction. So $a^s \neq a^r b$ for all r, s , and we have exhibited $2n$ distinct symmetries.

There are no others. We may assume the centre O of π_n is the origin, so every symmetry φ fixes O and hence is a linear transformation ([Proposition 1.2.29](#)). The vertices adjacent to v_0 , namely v_1 and v_{n-1} , are the closest vertices to v_0 ; that is, $\|v_i - v_0\| > \|v_1 - v_0\|$ for $2 \leq i \leq n-2$. So if $\varphi(v_0) = v_j$, then $\varphi(v_1) = v_{j+1}$ or $\varphi(v_1) = v_{j-1}$. In the first case a^j agrees with φ on v_0 and v_1 , so [Corollary 1.2.32](#) gives $\varphi = a^j$; in the second case $\varphi = a^j b$ for the same reason. Therefore $|\Sigma(\pi_n)| = 2n$.

Finally, $bab = a^{-1}$: by [Corollary 1.2.32](#) it suffices to evaluate both sides on v_0 and v_1 , and $bab(v_0) = v_{n-1} = a^{-1}(v_0)$ while $bab(v_1) = v_0 = a^{-1}(v_1)$. ■

Note 1.2.40 (Symmetry in calculus). Symmetry arises in calculus when describing curves $f(x, y) = 0$:

- (i) symmetry about the x -axis: the equation is unaltered when y is replaced by $-y$ — in our language, ρ_x ;

- (ii) symmetry about the y -axis: x replaced by $-x$ — ρ_y ;
- (iii) symmetry with respect to the origin: both replaced — R_{180} ;
- (iv) symmetry about the line $y = x$: x and y interchanged — ρ_L with L the 45° line.

Thus $f(x, y) = f(x, -y)$ says the graph Γ of $f(x, y) = 0$ is symmetric about the x -axis, for $(a, b) \in \Gamma$ implies $(a, -b) \in \Gamma$.

§1.3. Subgroups and Lagrange's Theorem

A subgroup of a group G is a subset which is a group under the same operation as in G . The following definition makes this precise.

Definition 1.3.1 (Closed subset). Let $*$ be an operation on a set G and let $S \subseteq G$. We say S is **closed** under $*$ if $x * y \in S$ for all $x, y \in S$.

The operation on G is a function $*$: $G \times G \rightarrow G$; if $S \subseteq G$, then $S \times S \subseteq G \times G$, and S being closed means $*(S \times S) \subseteq S$. For example, $\mathbb{Z}$ is a subset of the additive group $\mathbb{Q}$ closed under $+$. However $\mathbb{Q}^\times$ is closed under multiplication but not under $+$: both 2 and -2 lie in $\mathbb{Q}^\times$, yet $-2 + 2 = 0 \notin \mathbb{Q}^\times$.

Definition 1.3.2 (Subgroup). A subset H of a group G is a **subgroup** if

- (i) $1 \in H$;
- (ii) if $x, y \in H$, then $xy \in H$ (that is, H is closed under $*$);
- (iii) if $x \in H$, then $x^{-1} \in H$.

Notation . We write $H \leq G$ to denote that H is a subgroup of G . Both $\{1\}$ and G are always subgroups of G . A subgroup H is **proper** if $H \neq G$, written $H < G$, and **nontrivial** if $H \neq \{1\}$.

Proposition 1.3.3

Every subgroup $H \leq G$ of a group G is itself a group.

Proof. Axiom (ii) shows H is closed under the operation of G , so H has an operation (the restriction of $*$ to $H \times H$). This operation is associative, since $(xy)z = x(yz)$ holds for all $x, y, z \in G$, in particular for all $x, y, z \in H$. Axiom (i) gives the identity and axiom (iii) gives inverses. ■

Intuition . It is *quicker* to check that a subset is a subgroup than to verify the group axioms from scratch, because associativity is inherited and need not be verified again.

Example 1.3.4 (Subgroups). (i) $O_2(\mathbb{R})$, the isometries fixing the origin, is a subgroup of $\text{Isom}(\mathbb{R}^2)$. If $\Omega \subseteq \mathbb{R}^2$, then $\Sigma(\Omega)$ is also a subgroup of $\text{Isom}(\mathbb{R}^2)$; and if the centre of gravity of Ω is at the origin, then $\Sigma(\Omega) \leq O_2(\mathbb{R})$.

(ii) The four permutations

$$\mathbf{V} = \{(1), (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\}$$

form a group, because $\mathbf{V}$ is a subgroup of S_4 : $(1) \in \mathbf{V}$; $\alpha^2 = (1)$ for each $\alpha \in \mathbf{V}$, so $\alpha^{-1} = \alpha \in \mathbf{V}$; and the product of any two distinct elements of $\mathbf{V} - \{(1)\}$ is the third one. One calls $\mathbf{V}$ the **four-group**, or **Klein group** ($\mathbf{V}$ abbreviates *Vierergruppe*).

Note how much work this saves: verifying $a(bc) = (ab)c$ directly would involve $4^3 = 64$ equations.

(iii) If $\mathbb{R}^2$ is the plane as an additive abelian group, then any line L through the origin is a subgroup: choose a point (a, b) on L and note that L consists of all scalar multiples (ra, rb) .

One can shorten the list of items needed to verify that a subset is a subgroup.

Proposition 1.3.5

A subset H of a group G is a subgroup if and only if H is nonempty and, whenever $x, y \in H$, then $xy^{-1} \in H$.

Proof. If H is a subgroup, then $H \neq \emptyset$ since $1 \in H$; and if $x, y \in H$, then $y^{-1} \in H$ by (iii), so $xy^{-1} \in H$ by (ii).

Conversely, assume H satisfies the new condition. Since H is nonempty it contains some h ; taking $x = h = y$ gives $1 = hh^{-1} \in H$, so (i) holds. If $y \in H$, set $x = 1$ (legitimate now that $1 \in H$) to get $y^{-1} = 1y^{-1} \in H$, so (iii) holds. Finally $(y^{-1})^{-1} = y$ by Lemma 1.2.10, so if $x, y \in H$, then $y^{-1} \in H$ and $xy = x(y^{-1})^{-1} \in H$, giving (ii). ■

Remark 1.3.6. Since every subgroup contains 1, one may replace “ H is nonempty” by “ $1 \in H$ ”. If the operation in G is addition, the condition reads: H is a nonempty subset with $x, y \in H \Rightarrow x - y \in H$.

Note 1.3.7 (Historical). For Galois a group was just a subset H of S_n closed under composition. A. Cayley, in 1854, was the first to define an abstract group, mentioning associativity, inverses and identity explicitly.

Proposition 1.3.8

A nonempty subset H of a *finite* group G is a subgroup if and only if H is closed under the operation of G . In particular, a nonempty subset of S_n is a subgroup if and only if it is closed under composition.

Proof. Every subgroup is nonempty and closed, by axioms (i) and (ii).

Conversely, assume H is a nonempty closed subset, so axiom (ii) holds. Then H contains all powers of its elements: there is some $a \in H$, and $a^n \in H$ for all $n \geq 1$. As in Example 1.2.19, every element of G has finite order, so there is m with $a^m = 1$; hence $1 \in H$ and axiom (i) holds. Finally, if $h \in H$ and $h^m = 1$, then $h^{-1} = h^{m-1} \in H$ (since $hh^{m-1} = 1 = h^{m-1}h$), so axiom (iii) holds. ■

Remark 1.3.9. This proposition can be false for infinite G : the subset $\mathbb{N}$ of the additive group $\mathbb{Z}$ is closed under $+$ but is not a subgroup.

Example 1.3.10 (The alternating group). The subset of S_n consisting of all the *even* permutations is a subgroup, because it is closed under multiplication: even $\circ$ even = even ([Corollary 1.1.40](#)). This subgroup is called the **alternating group** on n letters and is denoted by A_n .

Note 1.3.11. The name comes from polynomials: if $f(x) = (x - u_1) \cdots (x - u_n)$, then $D = \prod_{i < j} (u_i - u_j)$ changes sign when one permutes the roots, since $\prod_{i < j} [\alpha(u_i) - \alpha(u_j)] = \pm D$. The sign *alternates* as various α are applied; it does not change for α in the alternating group.

§1.3.i. Cyclic Groups

Definition 1.3.12 (Cyclic subgroup, cyclic group). If G is a group and $a \in G$, write

$$\langle a \rangle = \{a^n : n \in \mathbb{Z}\} = \{\text{all powers of } a\};$$

$\langle a \rangle$ is called the **cyclic subgroup** of G generated by a . A group G is **cyclic** if $G = \langle a \rangle$ for some $a \in G$, and then a is called a **generator** of G .

It is easy to see $\langle a \rangle$ is a subgroup: $1 = a^0 \in \langle a \rangle$; $a^n a^m = a^{n+m} \in \langle a \rangle$; $a^{-1} \in \langle a \rangle$. By [Example 1.2.11](#)(vi), for every $n \geq 1$ the multiplicative group Γ_n of all n th roots of unity is cyclic, with the primitive n th root of unity $\zeta = e^{2\pi i/n}$ as a generator. A cyclic group can have several different generators; for instance $\langle a \rangle = \langle a^{-1} \rangle$.

Proposition 1.3.13

If $G = \langle a \rangle$ is a cyclic group of order n , then a^k is a generator of G if and only if $\gcd(k, n) = 1$.

Proof. If a^k is a generator, then $a \in \langle a^k \rangle$, so $a = a^{ks}$ for some s . Hence $a^{ks-1} = 1$, and [Lemma 1.2.21](#) gives $n \mid (ks - 1)$; that is, $ks - 1 = tn$ for some integer t , so $sk - tn = 1$ and $\gcd(k, n) = 1$.

Conversely, if $1 = sk + tn$, then $a = a^{sk+tn} = a^{sk}$ (because $a^{tn} = 1$), so $a \in \langle a^k \rangle$. Hence $G = \langle a \rangle \leq \langle a^k \rangle$, and so $G = \langle a^k \rangle$. ■

Corollary 1.3.14

The number of generators of a cyclic group of order n is $\phi(n)$, where ϕ is the Euler ϕ -function.

Proof. Immediate from [Proposition 1.3.13](#). ■

Proposition 1.3.15

Every subgroup S of a cyclic group $G = \langle a \rangle$ is itself cyclic. In fact a^m is a generator of S , where m is the smallest positive integer with $a^m \in S$.

Proof. We may assume $S \neq \{1\}$, the statement being obvious otherwise. Let $I = \{m \in \mathbb{Z} : a^m \in S\}$. Then $0 \in I$, since $a^0 = 1 \in S$; if $m, n \in I$, then $a^m a^{-n} = a^{m-n} \in S$, so $m - n \in I$; and if $m \in I$, $i \in \mathbb{Z}$, then $(a^m)^i = a^{im} \in S$, so $im \in I$. Since $S \neq \{1\}$ there is $a^q \in S$ with $a^q \neq 1$, so $q \in I$ and $I \neq \{0\}$. Hence, if k is the smallest positive integer in I , then $k \mid m$ for every $m \in I$.

We claim $\langle a^k \rangle = S$. Clearly $\langle a^k \rangle \leq S$. Conversely, take $s \in S$; then $s = a^m$ for some $m \in I$, and $m = k\ell$ for some ℓ , so $s = a^{k\ell} \in \langle a^k \rangle$. ■

Proposition 1.3.16

Let G be a finite group and let $a \in G$. Then the order of a is the number of elements in $\langle a \rangle$.

Proof. Since G is finite, there is $k \geq 1$ such that $1, a, a^2, \dots, a^{k-1}$ are k distinct elements while $1, a, \dots, a^k$ has a repetition; hence $a^k = a^i$ for some $0 \leq i < k$. If $i \geq 1$, then $a^{k-i} = 1$, contradicting the original list having no repetitions. Therefore $a^k = a^0 = 1$, and k is the order of a , being the smallest such positive integer.

Put $H = \{1, a, a^2, \dots, a^{k-1}\}$, so $|H| = k$; it suffices to show $H = \langle a \rangle$. Clearly $H \subseteq \langle a \rangle$. Conversely, given $a^i \in \langle a \rangle$, the division algorithm gives $i = qk + r$ with $0 \leq r < k$, and then $a^i = a^{qk+r} = (a^k)^q a^r = a^r \in H$. ■

Definition 1.3.17 (Order of a group). If G is a finite group, then the number of elements in G , denoted $|G|$, is called the **order** of G .

Abuse of Notation 1.3.18. The word “order” is used in two senses: the order of an element $a \in G$, and the order $|G|$ of a group. [Proposition 1.3.16](#) reconciles them: the order of a equals $|\langle a \rangle|$.

Proposition 1.3.19

A group G of order n is cyclic if and only if, for each divisor d of n , there is at most one cyclic subgroup of order d .

Proof. Suppose $G = \langle a \rangle$ is cyclic of order n . We claim $\langle a^{n/d} \rangle$ has order d . Clearly $(a^{n/d})^d = a^n = 1$, and d is smallest with this property: if $(a^{n/d})^r = 1$, then $n \mid (n/d)r$ by [Lemma 1.2.21](#), so $(n/d)r = ns$ for some integer s , giving $r = ds \geq d$.

For uniqueness, let $C \leq G$ have order d . By [Proposition 1.3.15](#), C is cyclic, say $C = \langle x \rangle$ with $x = a^m$ of order d . Then $1 = x^d = a^{md}$, so $n \mid md$ by [Lemma 1.2.21](#), say $md = nk$. Therefore $x = a^m = (a^{n/d})^k$, so $C = \langle x \rangle \subseteq \langle a^{n/d} \rangle$; since both subgroups have order d , $C = \langle a^{n/d} \rangle$.

Conversely, define a relation on G by $a \equiv b$ if $\langle a \rangle = \langle b \rangle$. This is an equivalence relation, and the class $[a]$ consists of all the generators of $C = \langle a \rangle$; denote it by $\text{gen}(C)$. Then

$$G = \bigcup_{C \text{ cyclic}} \text{gen}(C), \quad \text{so} \quad n = |G| = \sum_C |\text{gen}(C)|,$$

the sum being over all cyclic subgroups of G . But $|\text{gen}(C)| = \phi(|C|)$ by [Corollary 1.3.14](#). By hypothesis G has at most one cyclic subgroup of any given order, so

$$n = \sum_C |\text{gen}(C)| \leq \sum_{d|n} \phi(d) = n.$$

Therefore, for each divisor d of n , there must be a cyclic subgroup C of order d contributing $\phi(d)$ to the sum. In particular there is a cyclic subgroup of order n , and so G is cyclic. ■

§1.3.ii. Subgroups Generated by a Subset**Proposition 1.3.20**

The intersection $\bigcap_{i \in I} H_i$ of any family of subgroups of a group G is again a subgroup of G . In particular, if $H, K \leq G$, then $H \cap K \leq G$.

Proof. Let $D = \bigcap_{i \in I} H_i$. Then $D \neq \emptyset$ because $1 \in H_i$ for all i . If $x \in D$, then $x \in H_i$ for all i ; each H_i is a subgroup, so $x^{-1} \in H_i$ for all i and $x^{-1} \in D$. Finally, if $x, y \in D$, then x, y lie in every H_i , hence so does xy , and $xy \in D$. ■

Corollary 1.3.21

If X is a subset of a group G , then there is a subgroup $\langle X \rangle$ of G containing X that is smallest in the sense that $\langle X \rangle \leq H$ for every subgroup $H \leq G$ with $X \subseteq H$.

Proof. There do exist subgroups of G containing X ; for example G itself. Define $\langle X \rangle = \bigcap_{X \subseteq H} H$, the intersection of all subgroups H of G containing X . By Proposition 1.3.20 this is a subgroup, and it contains X because each H does. If H is any subgroup containing X , then H is one of the subgroups being intersected, so $\langle X \rangle \leq H$. ■

Note 1.3.22. There is no restriction on X ; in particular $X = \emptyset$ is allowed. Since $\emptyset \subseteq H$ for every subgroup H , the subgroup $\langle \emptyset \rangle$ is the intersection of *all* subgroups of G , one of which is $\{1\}$; hence $\langle \emptyset \rangle = \{1\}$.

Definition 1.3.23 (Subgroup generated by a subset). If X is a subset of a group G , then $\langle X \rangle$ is called the **subgroup generated by X** .

Example 1.3.24. (i) If $G = \langle a \rangle$ is cyclic with generator a , then G is generated by $X = \{a\}$.

(ii) The symmetry group $\Sigma(\pi_n)$ of a regular n -gon is generated by a, b , where a is the rotation about the origin by $(360/n)^\circ$ and b is a reflection (Theorem 1.2.39). These generators satisfy $a^n = 1$, $b^2 = 1$ and $bab = a^{-1}$, so $\Sigma(\pi_n)$ is a dihedral group D_{2n} .

Definition 1.3.25 (Word). Let X be a nonempty subset of a group G . A **word** on X is either the identity element or an element of G of the form

$$w = x_1^{e_1} x_2^{e_2} \cdots x_n^{e_n},$$

where $n \geq 1$, $x_i \in X$ for all i , and $e_i = \pm 1$ for all i .

Proposition 1.3.26

If X is a nonempty subset of a group G , then $\langle X \rangle$ is the set of all words on X .

Proof. Let W be the set of all words on X ; we first show $W \leq G$. By definition $1 \in W$. If $w = x_1^{e_1} \cdots x_n^{e_n}$ and $w' = y_1^{f_1} \cdots y_m^{f_m}$ lie in W , then $ww' = x_1^{e_1} \cdots x_n^{e_n} y_1^{f_1} \cdots y_m^{f_m}$ is again a word on X , so $ww' \in W$; and $w^{-1} = x_n^{-e_n} x_{n-1}^{-e_{n-1}} \cdots x_1^{-e_1} \in W$. Thus W is a subgroup containing every element of X , and so $\langle X \rangle \leq W$.

For the reverse inequality, let S be any subgroup containing X . Since S is a subgroup it contains x^e whenever $x \in X$ and $e = \pm 1$, and it contains all products of such elements; hence $W \leq S$. Therefore $W \leq \bigcap S = \langle X \rangle$. ■

Proposition 1.3.27 (gcd and lcm, group-theoretically)

Let a and b be integers and let $A = \langle a \rangle$ and $B = \langle b \rangle$ be the cyclic subgroups of $\mathbb{Z}$ they generate.

(i) If $A + B = \{\alpha + \beta : \alpha \in A, \beta \in B\}$, then $A + B = \langle d \rangle$, where $d = \gcd(a, b)$.

(ii) $A \cap B = \langle m \rangle$, where $m = \text{lcm}(a, b)$.

Proof. (i) It is straightforward to check that $A + B$ is a subgroup of $\mathbb{Z}$; in fact $A + B$ is precisely the set of all linear combinations of a and b . By Proposition 1.3.15 the subgroup $A + B$ is cyclic: $A + B = \langle d \rangle$, where d can be chosen to be the smallest nonnegative number in $A + B$. Thus d is a common divisor of a and b , and it is the smallest such; that is, $d = \gcd(a, b)$.

(ii) If $c \in A \cap B$, then $a \mid c$ and $b \mid c$, so every element of $A \cap B$ is a common multiple of a and b ; conversely every common multiple lies in the intersection. By Proposition 1.3.15, $A \cap B = \langle m \rangle$, where m may be taken to be the smallest nonnegative number in $A \cap B$; that is, $m = \text{lcm}(a, b)$. ■

§1.3.iii. Cosets and Lagrange's Theorem

Perhaps the most fundamental fact about subgroups H of a finite group G is that their orders are constrained. Certainly $|H| \leq |G|$, but it turns out that $|H|$ must *divide* $|G|$. To prove this we introduce cosets.

Definition 1.3.28 (Coset). If $H \leq G$ and $a \in G$, then the **coset** aH is the subset

$$aH = \{ah : h \in H\}.$$

Of course $a = a1 \in aH$. Cosets are usually *not* subgroups: if $a \notin H$, then $1 \notin aH$, for $1 = ah$ would give the contradiction $a = h^{-1} \in H$. In $*$ notation we write $a * H = \{a * h : h \in H\}$; in additive notation, $a + H = \{a + h : h \in H\}$.

Note 1.3.29. The cosets just defined are often called *left* cosets; there are also *right* cosets $Ha = \{ha : h \in H\}$. We shall work almost exclusively with left cosets.

Example 1.3.30 (Cosets). (i) Consider $\mathbb{R}^2$ as an additive abelian group and let ℓ be a line through the origin O ; as in Example 1.3.4(iii), ℓ is a subgroup. If $\beta \in \mathbb{R}^2$, then the coset $\beta + \ell$ is the line ℓ' through β parallel to ℓ , for if $r\alpha \in \ell$, then the parallelogram law gives $\beta + r\alpha \in \ell'$ (see Figure 1.4).

(ii) If $G = S_3$ and $H = \langle (1\ 2) \rangle$, there are exactly three cosets of H , each of size 2:

$$\begin{aligned} H &= \{(1), (1\ 2)\} = (1\ 2)H, \\ (1\ 3)H &= \{(1\ 3), (1\ 2\ 3)\} = (1\ 2\ 3)H, \\ (2\ 3)H &= \{(2\ 3), (1\ 3\ 2)\} = (1\ 3\ 2)H. \end{aligned}$$

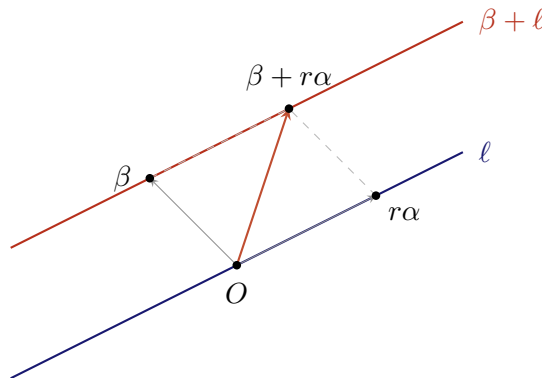


Figure 1.4.: The coset $\beta + \ell$ is the line through β parallel to ℓ .

Observation . In these examples different cosets of a given subgroup do not overlap. That is no accident: cosets are equivalence classes.

Proposition 1.3.31

If $H \leq G$, then the relation on G defined by

$$a \equiv b \quad \text{if} \quad a^{-1}b \in H$$

is an equivalence relation, and the equivalence class of $a \in G$ is the coset aH .

Proof. *Reflexive:* $a^{-1}a = 1 \in H$. *Symmetric:* if $a^{-1}b \in H$, then $(a^{-1}b)^{-1} = b^{-1}a \in H$, since subgroups are closed under inverses. *Transitive:* if $a^{-1}b, b^{-1}c \in H$, then $(a^{-1}b)(b^{-1}c) = a^{-1}c \in H$, since subgroups are closed under multiplication.

For the classes: if $x \equiv a$, then $x^{-1}a \in H$, and [Lemma 1.3.32](#) gives $x \in xH = aH$; thus $[a] \subseteq aH$. Conversely, if $x = ah \in aH$, then $x^{-1}a = (ah)^{-1}a = h^{-1}a^{-1}a = h^{-1} \in H$, so $x \equiv a$ and $x \in [a]$. Hence $[a] = aH$. ■

Lemma 1.3.32

Let $H \leq G$ and let $a, b \in G$.

- (i) $aH = bH$ if and only if $b^{-1}a \in H$. In particular, $aH = H$ if and only if $a \in H$.
- (ii) If $aH \cap bH \neq \emptyset$, then $aH = bH$.
- (iii) $|aH| = |H|$ for all $a \in G$.

Proof. (i) and (ii) are the general facts about equivalence classes applied to [Proposition 1.3.31](#): distinct classes are disjoint, and two elements are equivalent if and only if their classes coincide. The second statement of (i) follows because $H = 1H$.

(iii) The function $f: H \rightarrow aH$ given by $f(h) = ah$ is a bijection, with inverse $ah \mapsto a^{-1}(ah) = h$. Therefore H and aH have the same number of elements. ■

Theorem 1.3.33 (Lagrange's theorem)

If H is a subgroup of a finite group G , then $|H|$ is a divisor of $|G|$.

Proof. Let $\{a_1H, a_2H, \dots, a_tH\}$ be the family of all the distinct cosets of H in G . Then

$$G = a_1H \cup a_2H \cup \dots \cup a_tH,$$

because each $g \in G$ lies in the coset gH , and $gH = a_iH$ for some i . Moreover distinct cosets a_iH and a_jH are disjoint, by [Lemma 1.3.32\(ii\)](#). It follows that

$$|G| = |a_1H| + |a_2H| + \dots + |a_tH|.$$

But $|a_iH| = |H|$ for all i , by [Lemma 1.3.32\(iii\)](#), and so $|G| = t|H|$. ■

Definition 1.3.34 (Index). The **index** of a subgroup H in G , denoted $[G : H]$, is the number of cosets of H in G .

When G is finite, the index $[G : H]$ is the number t in the formula $|G| = t|H|$ above, so that

$$|G| = [G : H] |H|.$$

In particular the index $[G : H]$ is also a divisor of $|G|$.

Corollary 1.3.35

If H is a subgroup of a finite group G , then $[G : H] = |G|/|H|$.

Proof. Immediate from Lagrange's theorem. ■

Example 1.3.36. By [Theorem 1.2.39](#), the symmetry group $\Sigma(\pi_n)$ of a regular n -gon is a dihedral group of order $2n$. It contains the cyclic subgroup $\langle a \rangle$ of order n generated by a rotation a , and $[\Sigma(\pi_n) : \langle a \rangle] = 2$. Thus there are two cosets: $\langle a \rangle$ and $b\langle a \rangle$, where b is any symmetry outside $\langle a \rangle$.

Corollary 1.3.37

If G is a finite group and $a \in G$, then the order of a is a divisor of $|G|$.

Proof. By [Proposition 1.3.16](#), the order of a equals the order of the subgroup $H = \langle a \rangle$; now apply Lagrange's theorem. ■

This explains why the orders of the elements of S_5 displayed in [Table 1.2](#) are divisors of 120.

Corollary 1.3.38

If a finite group G has order m , then $a^m = 1$ for all $a \in G$.

Proof. By [Corollary 1.3.37](#), a has order d with $d \mid m$, say $m = dk$. Thus $a^m = a^{dk} = (a^d)^k = 1$. ■

Corollary 1.3.39

If p is a prime, then every group G of order p is cyclic.

Proof. Choose $a \in G$ with $a \neq 1$ and let $H = \langle a \rangle$. By Lagrange's theorem $|H|$ divides $|G| = p$; since p is prime and $|H| > 1$, it follows that $|H| = p = |G|$, and so $H = G$. ■

Question 1.3.40 (Is the converse of Lagrange's theorem true?). Lagrange's theorem says the order of a subgroup of a finite group G divides $|G|$. If d divides $|G|$, must there exist a subgroup of G of order d ?

Remark 1.3.41. The answer is *no*: the alternating group A_4 has order 12 but has no subgroup of order 6.

Lecture 2: Residue Classes and Units Modulo n

§1.4. Residue classes

18 September
2026

Fix an integer $n \geq 2$. Under addition, the set

$$n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$$

is a subgroup of $\mathbb{Z}$. Its cosets are the basic objects of this lecture.

Definition 1.4.1 (Residue class). For $a \in \mathbb{Z}$, the **residue class of a modulo n** is the coset

$$[a]_n = a + n\mathbb{Z} = \{a + kn : k \in \mathbb{Z}\}.$$

The set of all residue classes modulo n is denoted by

$$\mathbb{Z}_n = \{[a]_n : a \in \mathbb{Z}\}.$$

Proposition 1.4.2 (Equality of residue classes)

For $a, b \in \mathbb{Z}$, the following conditions are equivalent:

$$[a]_n = [b]_n, \quad a - b \in n\mathbb{Z}, \quad n \mid (a - b).$$

Proof. By the criterion for equality of cosets, $a + n\mathbb{Z} = b + n\mathbb{Z}$ if and only if $a - b \in n\mathbb{Z}$. The latter means precisely that $a - b = kn$ for some $k \in \mathbb{Z}$, which is equivalent to $n \mid (a - b)$. ■

Thus $[a]_n = [b]_n$ exactly when $a \equiv b \pmod{n}$. In particular, changing an integer by a multiple of n does not change its residue class.

Proposition 1.4.3 (Division algorithm)

Given $a \in \mathbb{Z}$, there exist unique integers q, r such that

$$a = qn + r \quad \text{and} \quad 0 \leq r < n.$$

Proof. For existence, consider

$$S = \{a - kn : k \in \mathbb{Z}\} \cap \mathbb{N}_0.$$

The set S is nonempty: for example, if $m \geq |a|$, then $a - (-m)n = a + mn \geq 0$. By the well-ordering principle, S has a least element r . Thus $r = a - qn$ for some $q \in \mathbb{Z}$, so $a = qn + r$. If $r \geq n$, then

$$r - n = a - (q + 1)n \geq 0,$$

so $r - n \in S$, contradicting the minimality of r . Hence $0 \leq r < n$, equivalently $0 \leq r \leq n - 1$.

For uniqueness, suppose $a = qn + r = q'n + r'$, where $0 \leq r, r' < n$. Then

$$r - r' = (q' - q)n.$$

But $|r - r'| < n$, and the only multiple of n with absolute value strictly less than n is 0. Therefore $r = r'$, and then $q = q'$. ■

Corollary 1.4.4 (Standard representative)

Every residue class in $\mathbb{Z}_n$ contains exactly one integer r satisfying $0 \leq r \leq n - 1$.

Proof. If $a = qn + r$ is the division-algorithm decomposition of a , then $a - r = qn$, so $[a]_n = [r]_n$ by [Proposition 1.4.2](#). If both $r, r' \in \{0, \dots, n - 1\}$ represent the same class, then $n \mid (r - r')$ and $|r - r'| < n$; hence $r = r'$. ■

Corollary 1.4.5

The integers $0, 1, \dots, n - 1$ form a complete set of coset representatives for $n\mathbb{Z}$ in $\mathbb{Z}$. Consequently,

$$\mathbb{Z}_n = \{[0]_n, [1]_n, \dots, [n - 1]_n\}$$

and these n residue classes are pairwise distinct.

Proof. Existence and distinctness are exactly the two assertions of [Corollary 1.4.4](#). ■

§1.5. The additive group $\mathbb{Z}_n$

We would like to add residue classes by adding representatives. Since a class has many representatives, we must first check that the answer does not depend on the choices.

Lemma 1.5.1 (Addition is well-defined)

The assignment

$$([a]_n, [b]_n) \mapsto [a + b]_n$$

is well-defined.

Proof. Suppose $[a_1]_n = [a_2]_n$ and $[b_1]_n = [b_2]_n$. Then $n \mid (a_1 - a_2)$ and $n \mid (b_1 - b_2)$. Therefore

$$n \mid (a_1 - a_2) + (b_1 - b_2) = (a_1 + b_1) - (a_2 + b_2),$$

which gives $[a_1 + b_1]_n = [a_2 + b_2]_n$. Thus different choices of representatives produce the same residue class. ■

Define addition on $\mathbb{Z}_n$ by

$$[a]_n + [b]_n = [a + b]_n.$$

Theorem 1.5.2

The set $\mathbb{Z}_n$ is an abelian group of order n under addition.

Proof. The operation is well-defined by [Lemma 1.5.1](#), and its value is again an element of $\mathbb{Z}_n$. For $[a]_n, [b]_n, [c]_n \in \mathbb{Z}_n$, associativity follows from associativity in $\mathbb{Z}$:

$$([a]_n + [b]_n) + [c]_n = [(a + b) + c]_n = [a + (b + c)]_n = [a]_n + ([b]_n + [c]_n).$$

The identity is $[0]_n$, because $[0]_n + [a]_n = [a]_n = [a]_n + [0]_n$. The inverse of $[a]_n$ is $[-a]_n$, since $[a]_n + [-a]_n = [0]_n$. Finally,

$$[a]_n + [b]_n = [a + b]_n = [b + a]_n = [b]_n + [a]_n,$$

so the group is abelian. By [Corollary 1.4.5](#), it has exactly n elements. ■

Example 1.5.3 (Subgroups of $\mathbb{Z}_{12}$). For $[a]_{12} \in \mathbb{Z}_{12}$, write

$$H_a = \langle [a]_{12} \rangle = \{[ka]_{12} : k \in \mathbb{Z}\}.$$

The distinct possibilities are

$$\begin{aligned} H_0 &= \{[0]_{12}\}, \\ H_6 &= \{[0]_{12}, [6]_{12}\}, \\ H_4 &= H_8 = \{[0]_{12}, [4]_{12}, [8]_{12}\}, \\ H_3 &= H_9 = \{[0]_{12}, [3]_{12}, [6]_{12}, [9]_{12}\}, \\ H_2 &= H_{10} = \{[0]_{12}, [2]_{12}, [4]_{12}, [6]_{12}, [8]_{12}, [10]_{12}\}, \\ H_1 &= H_5 = H_7 = H_{11} = \mathbb{Z}_{12}. \end{aligned}$$

For example, repeatedly adding $[5]_{12}$ gives $[0]_{12}, [5]_{12}, [10]_{12}, [3]_{12}, \dots$, and eventually every residue class occurs. Notice that 1, 5, 7, 11 are precisely the standard representatives that are relatively prime to 12.

The subgroup sizes in this example are governed by a general formula, proved in [Proposition 1.6.6](#) after we establish Bézout's identity.

§1.6. Multiplication and the group of units

Lemma 1.6.1 (Multiplication is well-defined)

The assignment

$$([a]_n, [b]_n) \mapsto [ab]_n$$

is well-defined.

Proof. Suppose $[a_1]_n = [a_2]_n$ and $[b_1]_n = [b_2]_n$. Write $a_1 - a_2 = kn$ and $b_1 - b_2 = \ell n$. Then

$$a_1 b_1 - a_2 b_2 = a_1(b_1 - b_2) + b_2(a_1 - a_2) = n(a_1 \ell + b_2 k).$$

Thus $n \mid (a_1 b_1 - a_2 b_2)$, so $[a_1 b_1]_n = [a_2 b_2]_n$. ■

We may therefore define multiplication on $\mathbb{Z}_n$ by

$$[a]_n [b]_n = [ab]_n.$$

Not every nonzero residue class has a multiplicative inverse. The invertible classes are characterized by the greatest common divisor.

Lemma 1.6.2 (The gcd is constant on residue classes)

If $[a]_n = [b]_n$, then $\gcd(a, n) = \gcd(b, n)$.

Proof. We have $a = b + kn$ for some $k \in \mathbb{Z}$. An integer divides both a and n if and only if it divides both $a - kn = b$ and n . Hence the pairs (a, n) and (b, n) have exactly the same common divisors, and therefore the same greatest common divisor. ■

Definition 1.6.3 (Units modulo n). The set of **units modulo n** is

$$\mathbb{Z}_n^* = \{[a]_n \in \mathbb{Z}_n : \gcd(a, n) = 1\}.$$

This definition does not depend on the representative a , by [Lemma 1.6.2](#).

The following result supplies the inverse needed to make $\mathbb{Z}_n^*$ a group.

Proposition 1.6.4 (Bézout's identity)

Let $a, b \in \mathbb{Z}$, not both zero, and let d be the least positive integer in

$$S = \{ax + by : x, y \in \mathbb{Z}\}.$$

Then $d = \gcd(a, b)$. In particular, there exist $x, y \in \mathbb{Z}$ such that

$$\gcd(a, b) = ax + by.$$

Proof. First, S contains a positive integer: one of a, b is nonzero, and either that integer or its negative is positive. Hence the least positive element d exists by the well-ordering principle. Write $d = ax_0 + by_0$.

We claim that $d \mid a$. By the division algorithm, write $a = qd + r$ with $0 \leq r < d$. Then

$$r = a - qd = a - q(ax_0 + by_0) = a(1 - qx_0) + b(-qy_0),$$

so $r \in S$. If $r > 0$, this contradicts the minimality of d ; hence $r = 0$ and $d \mid a$. The same argument gives $d \mid b$, so d is a common divisor of a and b .

Conversely, if c is any common divisor of a and b , then $c \mid ax_0 + by_0 = d$. Thus every common divisor of a, b divides d . Since $d > 0$ is itself a common divisor, it is the greatest positive common divisor: $d = \gcd(a, b)$. ■

Corollary 1.6.5 (Bézout criterion)

For integers a, b , one has $\gcd(a, b) = 1$ if and only if there exist $x, y \in \mathbb{Z}$ such that $ax + by = 1$.

Proof. If $\gcd(a, b) = 1$, the required integers exist by [Proposition 1.6.4](#). Conversely, if $ax + by = 1$, every common divisor of a and b divides 1; hence their greatest common divisor is 1. ■

Proposition 1.6.6 (Order of a residue class)

In the additive group $\mathbb{Z}_n$,

$$|\langle [a]_n \rangle| = \frac{n}{\gcd(a, n)}.$$

Consequently, $[a]_n$ generates $\mathbb{Z}_n$ if and only if $\gcd(a, n) = 1$.

Proof. Let $d = \gcd(a, n)$, and write $a = da'$ and $n = dn'$. Then $\gcd(a', n') = 1$. For a positive integer k ,

$$k[a]_n = [0]_n \Leftrightarrow n \mid ka \Leftrightarrow dn' \mid kda' \Leftrightarrow n' \mid ka'.$$

Choose $x, y \in \mathbb{Z}$ with $a'x + n'y = 1$. If $n' \mid ka'$, then

$$k = k(a'x + n'y) = (ka')x + n'(ky),$$

so $n' \mid k$. Conversely, $n' \mid k$ plainly implies $n' \mid ka'$. Therefore the least positive k for which $k[a]_n = [0]_n$ is $n' = n/d$. This is the order of $[a]_n$, and hence the size of the cyclic subgroup it generates. The final assertion follows because this size is n exactly when $d = 1$. ■

Corollary 1.6.7 (Criterion for invertibility)

A residue class $[a]_n \in \mathbb{Z}_n$ has a multiplicative inverse if and only if $\gcd(a, n) = 1$.

Proof. If $\gcd(a, n) = 1$, choose $x, y \in \mathbb{Z}$ with $ax + ny = 1$. Reducing this equality modulo n gives $[a]_n[x]_n = [ax]_n = [1]_n$, so $[x]_n$ is an inverse of $[a]_n$.

Conversely, suppose $[a]_n[x]_n = [1]_n$ for some $x \in \mathbb{Z}$. Then $ax \equiv 1 \pmod{n}$, so $ax - 1 = kn$ for some $k \in \mathbb{Z}$. Therefore $ax + n(-k) = 1$, and [Corollary 1.6.5](#) gives $\gcd(a, n) = 1$. ■

Theorem 1.6.8

The set $\mathbb{Z}_n^*$ is an abelian group under multiplication.

Proof. Multiplication is well-defined by [Lemma 1.6.1](#). If $[a]_n, [b]_n \in \mathbb{Z}_n^*$, then $\gcd(a, n) = \gcd(b, n) = 1$. Choose Bézout relations $ax + ny = 1$ and $bu + nv = 1$. Multiplying them gives

$$1 = (ax + ny)(bu + nv) = ab(xu) + n(axv + buy + nyv),$$

so $\gcd(ab, n) = 1$ by [Corollary 1.6.5](#). Hence $[ab]_n$ is again in $\mathbb{Z}_n^*$, proving closure.

Associativity and commutativity are inherited from integer multiplication, and $[1]_n$ is the identity. Every element of $\mathbb{Z}_n^*$ has an inverse by [Corollary 1.6.7](#); moreover that inverse also belongs to $\mathbb{Z}_n^*$, since it is itself invertible. Thus $\mathbb{Z}_n^*$ is an abelian group. ■

Example 1.6.9 (The units modulo 12). We have

$$\mathbb{Z}_{12}^* = \{[1]_{12}, [5]_{12}, [7]_{12}, [11]_{12}\}.$$

Each nonidentity element is its own inverse: $5^2 \equiv 7^2 \equiv 11^2 \equiv 1 \pmod{12}$. Thus $\mathbb{Z}_{12}^*$ is a group of order 4, but it is not cyclic.